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Wearable Urinary Collection Apparatus

Abstract

A wearable urinary collection system include a sleeve and a tubular membrane. The sleeve is configured to extend around at least a portion of a penis of a user. The tubular membrane is configured to collect urinary fluid within a volume. The tubular membrane may include a redundant fluid collection portion configured to extend around at least a portion of the sleeve and may also include a series of seals each extending along a width of the tubular membrane. The series of seals may be spaced along a length of the tubular membrane that is perpendicular to the width. Each of the seals may include an aperture. A guide tube may be configured to extend through the aperture of some of the seals to form a fluid flow path through the guide tube and the aperture of remaining ones of the seals into the volume.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation-in-part of U.S. application Ser. No. 19/042,647, filed on Jan. 31, 2025, which is a continuation-in-part of U.S. application Ser. No. 18/107,726, filed on Feb. 9, 2023, which claims priority to and the benefit of U.S. Provisional Application No. 63/372,241, filed on Feb. 28, 2022. U.S. application Ser. No. 19/042,647, filed on Jan. 31, 2025, also claims priority to and the benefit of U.S. Provisional Application No. 63/732,736, filed on Sep. 17, 2024. The contents of the foregoing applications are incorporated herein by reference in their entireties for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to wearable urinary collection apparatuses, and more particularly, to ambulatory and stationary urinary collection apparatuses configured to inhibit backflow of urinary fluid collected therein.

BACKGROUND

[0003] There are many known apparatus for collecting voided urine resulting from incontinency. One type of the known apparatus is wearable adult size diapers having materials which absorb urine, and which are typically intended to be disposed of after a single use. Another type of known apparatus includes some type of storage receptacle in which an individual's discharged urine is collected and stored, such that it may be disposed of (e.g., discharged) when there is an opportunity to do so. The present disclosure pertains to improvements to this later type of the known apparatus.

[0004] The present inventor has previously proposed several different types of wearable urinary collection apparatuses having a urine collection receptacle, including those discussed in U.S. Pat. No. 11,224,535 to Joh (the '535 Patent), the entire disclosure of which is incorporated herein by reference.

SUMMARY

[0005] One aspect of the present disclosure is a wearable urinary collection apparatus. The wearable urinary collection apparatus includes an elongate, tubular, flexible, collection receptacle which receives and stores urine therein. The collection receptacle also includes an entrance opening near one end thereof which is configured to receive a user's penis therethrough so that urine discharged from the user's penis will flow into the collection receptacle.

[0006] The collection receptacle also includes a constriction formed at an intermediate position of the collection receptacle. The constriction defines a constricted opening at the intermediate position that has a diameter smaller than an inner diameter of the collection receptacle. Accordingly, the urine which a user discharges into the collection receptacle must pass through the constricted opening to pass further inward of the collection receptacle from the constriction.

[0007] The wearable urinary collection apparatus also includes a tubular member that has opposite open ends and a diameter which is smaller than the inner diameter of the collection receptacle. The tubular member is shorter than the collection receptacle and has one of the opposite open ends thereof connected to the intermediate portion of the receptacle at the constriction in alignment with the constricted opening. Accordingly, the urine which passes through the constricted opening will pass into the tubular member.

[0008] The wearable urinary collection apparatus also includes a check valve provided at the one of the opposite open ends of the tubular member connected to the intermediate portion of the collection receptacle at the constriction. The check valve is configured to permit urine to freely

pass from the constricted opening into the tubular member. The check valve is also configured to block urine from flowing from the tubular member into the constricted opening.

[0009] In some implementations, each of the collection receptacle, the small tubular member and the check valve are constructed of plastic sheet material. Furthermore, the check valve may include two flaps of plastic sheet material, in which the two flaps are provided on opposite sides of the constricted opening. Each of the flaps may extend inward of the tubular member by an axial length which is longer than a sum of the diameter of the constricted opening and a distance from the constricted opening to the flap. Furthermore, each of the flaps may have one end separately secured at the constriction and the axial length of each of the flaps is longer than a sum of the diameter of the constricted opening and a distance from the constricted opening to where the flap is secured at the constriction. Additionally, each of the flaps may be spaced inward of the constriction. Finally, the axial length by which one of the flaps extends inward of the tubular member is longer than the axial length by which the other of the flaps extends inward of the tubular member.

[0010] In some implementations, the wearable urinary collection apparatus further includes an expandable entrance opening provided at the entrance opening near the one end of the collection receptacle. The expandable entrance opening may be configured to receive the user's penis therethrough. Furthermore, the expandable entrance opening may include an elastic band that normally urges the expandable entrance opening to engage the user's penis.

[0011] In some implementations, the wearable urinary collection apparatus further includes a harness which is configured to be disposed about a user's waist. The harness may include at least one fastener configured to secure the collection receptacle to the front portion of the harness such that the user may insert the user's penis into the entrance opening of the collection receptacle. The harness may include a waist strap, a pair of supporting straps, and a bridge connection. The waist strap is configured to be worn about a user's waist. The pair of supporting straps each have opposite ends connected to the waist strap so that the supporting straps extend on opposite sides of the user's pubic area. The bridge connection connects intermediate portions of the supporting straps at a level of the user's inferior pubic tubercle. The bridge connection may be 2-4 cm long and at least one fastener may be provided on at least one of the bridge connection portions of the supporting straps adjacent to the bridge connection.

[0012] In some implementations, the wearable urinary collection apparatus further includes a tubular supporting sleeve in which the collection receptacle may be disposed when the collection receptacle is being worn by the user. The tubular supporting sleeve may have an upper end which is open, and which is configured to surround the entrance opening of the collection receptacle when the collection receptacle is disposed in the tubular supporting sleeve. The tubular supporting sleeve may include at least one fastener at the upper end which is configured to secure the entrance opening of the tubular collection receptacle to the upper end. The at least one fastener at the first open end of the supporting sleeve may be configured to support the entrance opening of the tubular collection receptacle non-coaxially within the first open end of the supporting sleeve and to define an area of the first open end of the support sleeve adjacent to the entrance opening of the tubular collection receptacle through which the user may insert his scrotum into the tubular supporting sleeve. An opposite end of the tubular collection receptacle may have a closable opening provided therewith such that the opposite end of the tubular collection receptacle may be selectively opened or closed. The wearable urinary collection apparatus may further include an elongate, flexible member provided at the opposite end of the tubular collection receptacle which is configured to be gripped and moved by the user for manipulating the opposite end of the collection receptacle.

[0013] Another aspect of the present disclosure is a wearable urinary collection apparatus that includes an inner collection member, a urine collection receptacle, and a harness. The inner collection member is configured to receive at least a portion of a length of a penis of a user. The inner collection member extends from an upper end to a lower end. The inner collection member includes a voiding area disposed at the lower end of the inner collection member and a guide tube

extending from the voiding area away from the upper end. The urine collection receptacle is configured to collect urinary fluid therein. The urine collection receptacle includes one or more mating features, a redundant urine collection area configured to receive the inner collection member, and a tubular port disposed in the redundant urine collection area. The harness is configured to support the urine collection receptacle relative to the user. The harness includes securing straps configured to extend around respective portions of the user. The harness also includes one or more complementary mating features extending along a portion of at least one of the securing straps. The one or more complementary mating features are couplable to the one or more mating features of the urine collection receptacle. When the inner collection member is received in the redundant urine collection area of the urine collection receptacle, the guide tube extends through the tubular port into the urine collection receptacle to define a fluid flow path from the inner collection member to the urine collection receptacle.

[0014] Another aspect of the present disclosure is a urinary collection system that includes an inner member and a membrane. The inner member is configured to extend around at least a portion of a penis. The inner member includes a guide tube that extends from an end portion of the inner member. The membrane is configured to collect urinary fluid therein. The membrane includes a redundant fluid collection portion that is configured to extend around at least a portion of the inner member. The redundant fluid collection portion includes a port. The guide tube of the inner member is configured to extend through the port of the membrane to seal the membrane at an interface between the guide tube and the port and to form a fluid flow path from the inner member into the membrane. The redundant urine collection portion is configured to collect the urinary fluid that leaks through the seal formed at the interface between the guide tube and the port.

[0015] Another aspect of the present disclosure is a method for affixing a wearable urinary collection apparatus to a user having a penis. The method includes inserting a length of the penis into an inner collection member. The inner collection member includes a penile sleeve configured to extend along a shaft of the penis, and a stream tube configured to be adjacent to a terminal end of the penis. The stream tube includes a flexible valve disposed at a terminal end thereof. The method also include inserting the flexible valve of the inner collection member through a contractile tube of a urine collection receptacle to dispose the flexible valve within the urine collection receptacle and to form a fluid flow path from the inner collection member into the urine collection receptacle. The urine collection receptacle is configured to collect urine therein. An interface between the stream tube and the contractile tube is configured to seal the urine collection receptacle. The method also includes inserting the inner collection member into a redundant collection area of the urine collection receptacle. The redundant collection area is configured to collect the urine that leaks through the interface between the stream tube and the contractile tube. The redundant collection area includes the contractile tube.

[0016] Another aspect of the present disclosure is a wearable urinary collection apparatus that includes an inner member, a tubular membrane configured to collect urinary fluid therein, and a harness. The inner tube member includes a stream guide tube disposed at an end portion of the inner member, in which the end portion of the inner member is configured to receive a penis glands of a penis of a user. The inner member also includes a sleeve portion extending from the end portion of the inner member and configured to extend over a shaft of the penis of the user. The tubular membrane includes an inverted portion formed from an upper end of the tubular membrane inverting inwards at an inversion such that the tubular membrane extends longitudinally from a lower end of the tubular membrane that is opposite the upper end to terminate at the inversion. The tubular membrane also includes a contractile stem tube disposed within the inverted portion and extending through the tubular membrane. The inverted portion is configured to receive the inner member such that the stream guide tube extends through the contractile stem tube to provide a fluid flow path into the tubular membrane from the inner member. The tubular membrane also includes a mating feature extending away from the tubular membrane. The harness includes a complementary

mating feature releasably couplable to the mating feature of the tubular membrane. The harness is configured to support the membrane relative to the user such that the inversion of the tubular membrane is spaced from the user and the lower end of the tubular membrane extends away from the user.

[0017] Another aspect of the present disclosure is a urinary collection system that includes a sleeve and a tubular membrane. The sleeve is configured to extend around at least a portion of a penis of a user. The tubular membrane is configured to collect urinary fluid within a volume defined by the tubular membrane. The sleeve includes a guide tube that extends from an end of the sleeve. The tubular membrane may include a redundant fluid collection portion configured to extend around at least a portion of the sleeve. The tubular membrane may also include a series of seals each extending along a width of the tubular membrane. The series of seals may be spaced along a length of the tubular membrane that is perpendicular to the width, and each of the seals may include an aperture. The tubular membrane may also include one or more mating features. The wearable urinary collection system may also include a harness that is configured to be worn by the user. The harness may include a complementary mating feature that is couplable to the one or more mating features of the tubular membrane. The guide tube of the sleeve may be configured to extend through the aperture of some of the seals to form a fluid flow path from the sleeve through the guide tube and the aperture of remaining ones of the seals into the volume.

[0018] Another aspect of the present disclosure is a wearable urinary collection apparatus that includes an inner member and a fluid collection receptacle. The inner member may be configured to extend around a portion of a penis and includes an end portion configured to extend around a penis glans of the penis. The inner member may also include a tubular portion extending from the end portion. The fluid collection receptacle may include a backflow prevention region comprising a series of redundant collection chambers fluidly connected by respective apertures. The fluid collection receptacle may also include an upper region configured to extend around at least a portion of the inner member and fluidly connected to an upper redundant collection chamber from the redundant collection chambers via an upper aperture from the respective apertures. The fluid collection receptacle may also include a urinary collection region configured to collect fluid from the penis and fluidly connected to a lower redundant collection chamber from the redundant collection chambers via a lower aperture from the respective apertures. The tubular portion may be configured to extend through the upper redundant collection chamber via the upper aperture to terminate in the backflow prevention region such that the fluid from the penis is directed through the tubular portion and the lower aperture into the urinary collection region.

[0019] Another aspect of the present disclosure is a method for affixing a urinary collection system to a user having a penis. The method may include inserting a penis glans of the penis into a sleeve. The sleeve may include an end portion extending around the penis glans and a guide tube adjacent to a urethra of the penis. The method may also include inserting the sleeve into a redundant fluid collection portion of a urine collection receptacle. The urine collection receptacle may include a volume configured to collect urinary fluid and a series of sealed chambers disposed between the redundant fluid collection portion and fluidly connected by respective apertures. The method may also include inserting the guide tube of the sleeve through some of the sealed chambers via the respective apertures to form a fluid flow path from the sleeve through the guide tube and the respective apertures of remaining ones of the sealed chambers into the volume. The sealed chambers may be configured to collect the urinary fluid that travels through respective interfaces between the guide tube and the respective apertures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1A is a side cross-sectional view illustration of an example of a known urinary collection apparatus previously proposed by the present inventor that includes a tubular entrance port provided near an entrance opening of a urine collection receptacle.

[0021] FIG. 1B is a side cross-section view illustration of the known urinary collection apparatus of FIG. 1A that depicts connections between the urine collection receptacle and the entrance opening for assembly thereof.

[0022] FIG. 2A is a front view illustration of an example of a known harness previously proposed by the present inventor that is configured to be secured around a waist of a user to support the urinary collection apparatus of FIG. 1A adjacent to a penis of the user.

[0023] FIG. 2B includes rear view illustrations of examples of the known harness of FIG. 2A that depict a waist strap extending around the waist of the user and one or more securing straps extending from the waist strap around a buttocks of the user to support the urinary collection apparatus of FIG. 1A adjacent to the penis of the user.

[0024] FIG. 3A is an upper perspective view illustration of a known elastomeric cap previously proposed by the present inventor that is configured to be secured around the penis or a penis glans of the user such that when the penis glans is inserted into the entrance port of the urinary collection apparatus of FIG. 1A, the known elastomeric cap prevents urine that is being discharged into the entrance port from leaking therefrom without causing discomfort to the user.

[0025] FIG. 3B is a side cross-sectional view illustration of the known elastomeric cap of FIG. 3A showing baffles that may further prevent urine that is being discharged into the entrance port from leaking therefrom without causing discomfort to the user.

[0026] FIG. 4 is a side cross-sectional view illustration of an example of a urinary collection apparatus according to the present disclosure including a urine collection receptacle having backflow preventing means.

[0027] FIG. 5A is a side cross-sectional view illustration of the urinary collection apparatus of FIG. 4 showing a short tubular member of the backflow preventing means connected to the urine collection receptacle at a constriction formed at an intermediate portion of the urine collection receptacle and including a check valve formed from flaps.

[0028] FIG. 5B is a side cross-sectional view illustration of the urinary collection apparatus of FIG. 4 showing another implementation of the backflow preventing means in which the flaps are spaced inwardly of the short tubular member from the constriction.

[0029] FIG. 6 is a side view illustration of the short tubular member of FIG. 4 before being attached to the urine collection receptacle.

[0030] FIGS. 7A-7C are side view illustrations of the urinary collection apparatus of FIG. 4 demonstrating an example of a method for forming an expandable entrance opening one end of the urine collection receptacle according to the present disclosure, in which internal structures are shown using broken lines.

[0031] FIGS. 8-11 are side view illustrations of the urinary collection apparatus of FIG. 4 demonstrating a method for assembling the backflow preventing means with the urine collection receptacle according to the present disclosure.

[0032] FIG. 12 is an upper perspective view illustration of an example of a harness according to the present disclosure that is configured to support the urine collection receptacle of the FIG. 4 adjacent to the penis of the user.

[0033] FIG. 13 is an upper perspective view illustration of an example of a supporting sleeve according to the present disclosure that is configured to receive the urine collection receptacle of FIG. 4.

[0034] FIG. 14 is an upper perspective view illustration of the supporting sleeve of FIG. 13 showing the entrance opening of the urine collection receptacle of FIG. 4 being supported in a non-

coaxial manner within an entrance opening of the supporting sleeve to define a space for a user to insert his scrotum into the supporting sleeve.

[0035] FIG. **15** is an upper perspective view illustration of an example of a wearable urinary collection apparatus according to the present disclosure including an inner collection member disposed within a redundant urine collection area of a urine collection receptacle, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0036] FIGS. **16A-16C** are upper perspective view illustrations of the wearable urinary collection apparatus of FIG. **15** demonstrating an example of a method for forming the redundant urine collection area and inserting the inner collection member into the redundant urine collection area, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0037] FIG. **17** is an upper perspective view illustration of another example of the wearable urinary collection apparatus of FIG. **15** in which the urine collection receptacle includes a contractile ring, and in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0038] FIG. **18** is an upper perspective view illustration of another example of the wearable urinary collection apparatus of FIG. **15** in which the urine collection receptacle includes an outer adaptor ring connected to an inner opening ring, and in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0039] FIG. **19A** is an upper perspective view illustration of another example of the urine collection receptacle of FIG. **15** including an example of mating features for connection with elongate members, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0040] FIG. **19B** is an upper perspective view illustration of another example of the urine collection receptacle of FIG. **15** including another example of the mating features for connection with the elongate mating members, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0041] FIG. **20** is an upper perspective view illustration of another example of the urine collection receptacle of FIG. **15** demonstrating an example of a method for connecting examples of the urine collection receptacle and the redundant collection area to an example of the inner opening ring of FIG. **18**, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0042] FIG. **21A** is a perspective view illustration of an example of a harness according to the present disclosure including connecting feature for connection with the urine collection receptacle of FIG. **15**.

[0043] FIG. **21B** is a perspective view illustration of another example of the harness of FIG. **18A** including a waist strap.

[0044] FIG. **22** is a flow diagram of an example of a method for affixing the wearable urinary collection apparatus of FIG. **15** to a user having a penis.

[0045] FIG. **23** is an upper perspective view illustration of an example of a wearable urinary collection system according to the present disclosure including an inner member disposed within a redundant fluid collection portion of a urine collection receptacle that includes a backflow prevention region, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity.

[0046] FIGS. **24A-24B** are upper perspective view illustrations of the inner sleeve member of FIG. **23** in a folded and an unfolded configurations, respectively.

[0047] FIG. **25** is a flow diagram of an example of a method for affixing the wearable urinary collection system of FIG. **23** to a user having a penis.

[0048] FIG. **26** is an upper perspective view illustration of another example of a wearable urinary collection system according to the present disclosure including an inner sleeve and a guide tube

disposed within a redundant fluid collection portion of a urine collection receptacle, in which certain components are shown as hidden or transparent, indicated by broken lines, for clarity. [0049] FIG. 27 is an exploded view illustration of the wearable urinary collection system of FIG. 26.

[0050] FIG. 28 is an upper perspective view illustration of the wearable urinary collection system of FIG. 26 supported to a user by dermal elements spaced around a base of a penis of the user.

[0051] FIG. 29 is a flow diagram of an example of a method for affixing the wearable urinary collection system of FIG. 26 to a user having a penis.

DETAILED DESCRIPTION

[0052] Although the following disclosure offered for public dissemination is detailed to ensure adequacy and aid in understanding of the disclosed implementations, the following disclosure is not intended to prejudice that purpose of a patent which is to cover each new inventive concept therein no matter how it may later be disguised by variations in form or additions of further improvements. There have been chosen specific exemplary implementations of a urinary collection apparatus according to the present disclosure and specific alternative structures and modifications thereto. The exemplary implementations chosen for the purposes of illustration and description of the structures and methods described herein are shown in the accompanying drawings forming a part of the specification.

[0053] The advantages and features of the present disclosure will be readily apparent from the following detailed description. The present disclosure relates to a wearable urinary collection apparatus for use, for example, by an active individual with urinary incontinence issues. The wearable urinary collection apparatus described herein may be configured to be inconspicuously and comfortably worn by the individual to collect and safely store urine voided by the individual and may include means for preventing backflow and leakage. The wearable urinary collection apparatus described herein may be easy and cost efficient to manufacture. More particularly, the present disclosure pertains to such wearable urinary collection apparatuses which improve over conventional wearable urinary collection apparatuses, as well as methods of making the same.

[0054] As described previously, the present inventor has previously proposed several different types of wearable urinary collection apparatuses having a urine collection receptacle, including those discussed in U.S. Pat. No. 11,224,535 to Joh (the '535 Patent). Each of these previously proposed apparatuses generally include a lightweight, tubular collection receptacle formed of thin plastic sheet material which is configured to receive and store urine therein and some type of backflow restricting/preventing means integrally provided with the collection receptacle. Some previously proposed apparatus also include means for conveniently and inconspicuously supporting the collection receptacle on an individual's body adjacent to the individual's penis such that the individual may readily void urine into the receptacle whenever the individual has need to do so.

[0055] FIGS. 1A-3B of the present disclosure correspond to some of the drawings in the '535 patent and depict the components of one of the present inventor's previously proposed urinary collection apparatuses. FIG. 1A depicts an example of a known wearable urinary collection apparatus 101 previously proposed by the present inventor that includes an elongate, tubular, urine collection receptacle 100 which may be 30 to 80 cm long and 8 to 12 cm wide and which integrally includes a backflow preventing means provided with an elongate, tubular entrance port 102 that is attached within the urine collection receptacle 100 at or near one end of the urine collection receptacle 100. FIG. 1B depicts a manner in which the structure of FIG. 1A is produced. FIGS. 2A-2B depict a harness 104 which is configured to be secured around a waist of the user and to support the urine collection receptacle 100 adjacent to a penis of the user so that a penis glans thereof may be readily inserted into and/or remain in the entrance port while the urine collection receptacle 100 extends downward from the harness 104 so that urine discharged from the penis flows down through the entrance port into the urine collection receptacle 100 to be collected. Finally, FIGS. 3A-3B depict an elastomeric cap 110 that may be secured around the user's penis or penis glans

such that when the penis glans is inserted into the entrance port **102**, the elastomeric cap prevents urine that is being discharged into the entrance port **102** from leaking out without causing discomfort to the user. FIG. **1A** also depicts a lower portion of the urine collection receptacle **100** including a closable opening **107**, which may be used to selectively empty any urine which has been collected in the urine collection receptacle **100**, and an elongate cord **109** which is connected at a lower end of the urine collection receptacle **100**, which may be used by an individual for manipulating the urine collection receptacle **100** when desired. While the discussion herein focuses on users having a penis, '535 patent also discloses an adapter which permits the urinary collection apparatus to be used by users that do not have a penis (e.g., users who have female sexual organs). [0056] As disclosed in the '535 patent, the urine collection receptacle **100** and the entrance port **102** depicted in FIGS. **1A-1B** may be formed of plastic sheet material, e.g., any suitable type of plastic, plastic-like, rubber, elastomeric, or polymeric material(s), which material(s) may possess or be treated to possess anti-bacterial properties, including very thin sheet materials such as latex, thermoplastic polyurethane (TPU), various types of polyethylene including high density polyethylene (HDPE), low density polyethylene (LDPE), etc. TPU sheet material is more stretchable and elastic than polyethylene sheet material and may be more suitable for use in the present disclosure as the increased stretchability and elasticity make the urine collection receptacle **100** more resistant to tearing and leaking, e.g., so that the urine collection receptacle **100** or portions thereof may reliably expand as urine is flows therein or as the urine collection receptacle **100** is moved, sat upon, or manipulated. The urine collection receptacle **100** and the entrance port **102** may be relatively inexpensive to construct from such material(s), so that they may be disposed of after a single use/wearing but are sufficiently durable that they may be cleaned and reused if desired. By constructing these components of plastic sheet material, they are desirably lightweight and tend to be flat when empty.

[0057] Such plastic sheet material components and/or portions thereof may be readily bonded together using only heat or energy pulses. This is a convenient and inexpensive means for securing the entrance port **102** within the one end of the urine collection receptacle **100** without unduly constricting urine flow through either the entrance port **102** or the urine collection receptacle **100**.

[0058] According to an aspect of the known wearable urinary collection apparatus **101**, and as shown in FIGS. **1A-1B**, backflow preventing means are integrally provided with the entrance port **102** in the form of constrictions **106** that are formed by bonding opposing surfaces of the entrance port **102** together to significantly reduce the inner diameter (“ID”) of the entrance port **102** at one or more locations along its length as the entrance port **102** extends into the urine collection receptacle **100**, together with a so-called “flap” **108** which is an innermost portion of the entrance port **102** that extends inward of an innermost one of the constrictions **106**. Again, these constrictions may be conveniently and inexpensively formed using only heat or energy pulses and may reduce the ID of the entrance port by 50% or more. The depicted implementation includes four such constrictions **106**, which are aligned with each other at a center of the entrance port **102** in the width direction. However, as discussed in the '535 patent, the constrictions **106** may be provided in various numbers and in one or more shapes and sizes at various portions along the length of the entrance port **102**.

[0059] While the entrance port **102** is securely fixed within the urine collection receptacle **100** as depicted in FIGS. **1A-1B**, the innermost portion of the entrance port **102**, provided inward of an inner most one of the constrictions **106**, forms the flap **108**, described above, which may move to some extent relative to the urine collection receptacle **100**. As disclosed in the '535 patent, the flap **108** is particularly effective for preventing backflow of urine through the entrance port **102**, especially when the flap **108** is used in combination with the constrictions **106**. The opposing walls of the flap **108** will tend to remain engaged together other than when urine is flowing through the entrance port **102** into the main body and thereby prevents urine that has collected in the urine collection receptacle **100** from flowing back through the flap **108** even if some pressure is applied

to the urine collection receptacle **100** when a user is active, sitting, or lying down. The inventor has determined that width and length of the flap **108** are important for achieving optimum backflow prevention, while assuring the ability of urine discharged by a user to surely flow through the entrance port **102** into the urine collection receptacle **100**. For example, the inventor has determined that when the entrance port **102** including the flap **108** has a flat width of 8-9 cm and the flap **108** has an axial length of 4-6 cm extending inward of the urine collection receptacle **100** from the innermost one of the constrictions **106**, which limits the ID of the flow path at the constriction to 2.5-4.0 cm, which is effective for preventing backflow of urine through the flap **108** and the entrance port **102** while permitting urine to surely flow through the entrance port **102** into the urine collection receptacle **100**. Where the flap **108** length is shorter than 4.0 cm, it may not surely prevent backflow therethrough. Where the flap **108** length is more than 6.0 cm, it may simply require additional material without providing any better backflow prevention. The cord **109** shown in FIG. **1A** desirably permits a user to manipulate the lower portion of the urine collection receptacle **100** when desired to prevent the urine collection receptacle **100** from interfering with the user's movements, e.g., the user may move the lower portion of the urine collection receptacle **100** and any urine contained therein to avoid the user sitting or lying on the urine collection receptacle **100** and applying significant pressure to the urine collection receptacle **100**.

[0060] The harness **104** previously proposed by the inventor for supporting the urine collection receptacle **100** on a user's body is shown in FIGS. **2A-2B**, with FIG. **2A** depicting a perspective view of the harness **104** not attached to a user and FIG. **2B** depicting manners in which the harness **104** may be worn by a user. Generally, the harness **104** includes an adjustable waist strap **112** which may be secured around a user's waist, a padded member **114** which is configured to extend downward from the waist strap **112** in front of a user's genital area and to have the urine collection receptacle **100** secured thereto, and a pair of securing straps **116**, each of which has one end connected to a lower end of the padded member and an opposite free end which may be selectively secured to various portions of the waist strap **112** using an appropriate fastener, e.g., a hook-and-loop type fastener, a clip, a buckle, etc.

[0061] The waist strap **112** may be of any appropriate length, may include a fastener associated with free ends of the strap, e.g., a hook-and-loop type fastener, a clip, a buckle, etc., and may include means for adjusting the effective length of the waist strap **112**. The waist strap **112** may be formed of any appropriate material that will not cause any discomfort to the user even if the harness is worn by the user for an extended length of time. For example, the waist strap **112** may be made of leather, fabric, elastic material, cushioning material, or a combination of two or more of these.

[0062] The padded member **114** may be shaped like an inverted U and provided to extend downward from the waist strap **112** in front of a user's genital area, including a first portion that extends parallel along a portion of the waist strap and a pair of opposing arms that extend at right angles to the first portion, and may have a size appropriate for covering the user's genital area, e.g., the first portion may be 8-12 cm long and 2-4 cm wide, and the arms may be 5-7 cm long and 2-4 cm wide. The padded member **114** may be formed of padded or cushioned material(s) that will not cause any discomfort or allergic reaction to a user and may be washed for reuse, e.g., fabric(s), foam covered fabric(s), etc. Additionally, the padded member **114** may be provided with securing means **118** for securing a urine collection receptacle (e.g., the urine collection receptacle **100**) thereto, e.g., loops or hooks which are connected thereto and extend inward of the inverted U shape such that the entrance port **102** within the urine collection receptacle **100** may be conveniently disposed so that a user may readily insert his penis into the entrance port **102**. The urine collection receptacle **100** and/or the entrance port **102** may be provided with straps (not shown) that may be tied or otherwise secured to the securing means **118** such that the entrance openings of these components are disposed directly in front of the user's genital area and such that the user may readily insert his penis into the entrance port **102** when desired.

[0063] The securing straps **116** of the harness may each generally comprise an elongate strip of material having one end secured to an end of one of the arms of the padded member **114** and having a fastener provided on the opposite end of the securing straps **116** for securing the end of the securing straps **116** to the waist strap **112**, e.g., a hook-and-loop fastener, a clip, a button, etc., and intermediate portions of the securing straps **116** may be secured together with some type of fastener, e.g., thread which sews the straps together at the intermediate portions, a hook-and-loop fastener, a clip, etc. The material(s) used in forming the securing straps **116** may be the same as those used for making the waist strap **112** such as discussed above. As depicted in FIG. 2B, the securing straps **116** may be fastened to the waist strap **112** in various manners depending on the user's preference, e.g., the securing straps **116** may be arranged to fully overlap with each other and then may be extended between the user's legs adjacent to the user's genital area and having their free ends jointly fastened to the same portion of the waist strap **112**, the free ends of the securing straps **116** may be extended through user's legs adjacent to the user's genital area and separately secured to different portions of the waist strap **112**, etc.

[0064] Referring to FIGS. 3A-3B, FIG. 3A is a perspective view of the elastomeric cap **110** and FIG. 3B is a cross section of the elastomeric cap **110** along line B-B in FIG. 3A. The elastomeric cap **110** may be in the form of a cylindrical member constructed of soft flexible material such as a highly elastic polymer or polymer foam formed of silicone or other appropriate material which would be provided around a user's penis or penis glans so that when the penis glans is extended into the entrance port **102** the elastomeric cap **110** forms a leak-preventing seal with the inner surface of the entrance port **102**. The elastomeric cap **110** may be generally cylindrical in shape with relatively thick walls of soft elastomeric material, with a central opening **120** extending axially therethrough and opposite end surfaces of the elastomeric cap **110** extending concave inward of the elastomeric cap **110**. The diameter of the central opening **120** may be any appropriate size that will snugly engage the user's penis or penis glans when the cap is provided on the penis, but without causing any discomfort to the user. For example, the central opening **120** may have a diameter of 1-2 cm, noting that the highly elastic material of the elastomeric cap **110** will stretch to increase the size of the central opening **120** to fit the penis. If desired, a lubricant such as a hydrogel or the like may also be applied to surface(s) of the elastomeric cap **110**, which is inserted into the entrance port **102** of the urine collection receptacle **100** to help prevent undesired backflow and seepage of urine. The degree of the concave shape of the opposite end surfaces of the elastomeric cap **110** may be such that in cross section the two sides of the elastomeric cap **110** extending from the central opening **120** appear like opposing isosceles triangles. With this shape, the elastomeric cap **110** will only engage a small portion of the user's penis but will provide a much larger surface area for engagement with the inner surface of the entrance port **102** of the urine collection receptacle **100** for preventing backflow and seepage.

[0065] As shown in FIG. 3B, the elastomeric cap **110** may also include a baffle **122** provided in association with a discharge end of the central opening **120**. The baffle **122** may include flap(s) of a flexible plastic material, e.g., TPU, HDPE, LDPE, LLDPE, that extend from the end surface of the elastomeric cap **110** such that adjacent walls of the flaps will permit urine to flow downward therethrough, but which would otherwise remain engaged together or with the user's penis to prevent leakage of urine.

[0066] While such urinary collection apparatuses previously proposed by the present inventor functions appropriately, there remains a need in the art for the known devices to be improved on in several respects. For example, while some of the inventor's previously proposed apparatuses (e.g., the known wearable urinary collection apparatus **101** depicted in FIGS. 1A-3B) normally function very well to collect urine in the urine collection receptacle **100** and prevent urine from back flowing and/or otherwise leaking from the urine collection receptacle **100**, the inventor has determined through experimental use of the previously proposed implementations that there may be some situations where the known wearable urinary collection apparatus **101** does not sufficiently

prevent backflow and/or leakage of urine from the urine collection receptacle **100**, particularly when the user is engaged in activities or when the individual is sitting or lying down and the urine collection receptacle **100** contains a relatively large amount of urine, e.g., eight or more fluid ounces. Furthermore, the previously proposed backflow preventing means may also unduly inhibit urine which is being discharged into the apparatus from freely and reliably flowing into the urine collection receptacle **100** of the known wearable urinary collection apparatus **101**, which may result in leakage of the urine. Also, the previously proposed structures for supporting the urine collection receptacle **100** on an individual, including the harness **104** wearable about the individual's waist, could be improved in terms of reliability and convenience in use. Still further, the known wearable urinary collection apparatus **101** could be improved on in terms of conveniently preventing leakage of urine as it is being discharged into the urine collection receptacle **100** while also avoiding any discomfort to the individual.

[0067] The present inventor has carefully studied the known apparatuses, such as those disclosed in the '535 patent (e.g., the known wearable urinary collection apparatus **101**), and has performed substantial research regarding the discussed need and has discovered several improvements to the known apparatuses as discussed herein to arrive at an improved urinary collection apparatus. FIG. **4** is a cross-section view illustration of an example of a wearable urinary collection apparatus **1** according to the present disclosure. As shown, the wearable urinary collection apparatus **1** includes a urine collection receptacle **2** having backflow preventing means **4** disposed there-along. The wearable urinary collection apparatus **1** also includes an entrance opening **6** at one end thereof.

[0068] Referring to FIGS. **4-6**, the wearable urinary collection apparatus **1** is shown with integrated backflow prevention characteristics according to an exemplary implementation of the present disclosure. The wearable urinary collection apparatus **1** may generally comprise an elongate, tubular urine collection receptacle (e.g., the urine collection receptacle **2**) which may be formed with the expandable entrance opening **6** at one end thereof and an opposite end which may be closed or which may include a closable opening and/or a manipulation cord such as the cord **109** previously proposed by the inventor and shown in present FIG. **1A**, and a new, improved backflow preventing means (e.g., the backflow preventing means **4**) integrally provided with the urine collection receptacle **2**. The urine collection receptacle **2** may, for example, be 25-50 cm long and 8-15 cm wide, while the backflow preventing means **4** may include the smaller tubular member **8** which may be 6-9 cm long and 1.5-3 cm wide and connected within the urine collection receptacle **2** as discussed further herein. The wearable urinary collection apparatus **1** may also include the leakage preventing cap **60** which is simply fitted around the user's penis or penis glans, but not connected to the urine collection receptacle **2** or any other part of the wearable urinary collection apparatus **1**, as discussed further herein.

[0069] The urine collection receptacle **2** and the backflow preventing means **4** may be formed of plastic sheet material, e.g., any suitable type of plastic, plastic-like, rubber, elastomeric, or polymeric material(s), which material(s) may possess or be treated to possess anti-bacterial properties, including very thin sheet materials such as latex, thermoplastic polyurethane (TPU), various types of polyethylene including high density polyethylene (HDPE), low density polyethylene (LDPE), etc. TPU sheet material is more stretchable and elastic than polyethylene sheet material, and may be more suitable for use in the present disclosure as the increased stretchability and elasticity make the urine collection receptacle **2** more resistant to tearing, and leaking, e.g., so that the receptacle or portions thereof may reliably expand as urine is flowed into it or as the urine collection receptacle **2** is moved, sat upon, or manipulated. The urine collection receptacle **2** and the backflow preventing means **4** may be relatively inexpensive to construct from such material(s), so that they may be disposed of after a single use/wearing but are sufficiently durable that they may be cleaned and reused if desired. Such plastic sheet material components and/or portions thereof may be readily bonded together using only heat or energy pulses, which is a convenient and inexpensive manner of bonding components together and for selectively forming

constrictions in the components. By constructing these components of plastic sheet material they are desirably lightweight and tend to be flat when empty.

[0070] According to a first discovery, the inventor has determined that the entrance port **102** of the known wearable urinary collection apparatus **101**, which was formed as a separate, elongate tubular member having the backflow preventing means integrally formed therewith and secured within one end portion of the urine collection receptacle **100**, may not be necessary. Thus, the known wearable urinary collection apparatus **101** may be further simplified by eliminating the entrance port **102**. Instead of the entrance port **102**, the inventor has determined that a leakage inhibiting entrance opening—such as the expandable entrance opening **6**—configured for receipt of a user's penis may be directly formed or attached at the one end portion of the urine collection receptacle **2** while a simpler and more reliable backflow preventing means—such as the backflow preventing means **4**—may be provided integrally with the urine collection receptacle **2**.

[0071] According to aspects of the present disclosure, the inventor has determined that an expandable leakage inhibiting entrance opening (e.g., the expandable entrance opening **6**) for a user's penis or penis glans may be directly formed or attached at the one end portion of the urine collection receptacle **2** while the backflow preventing means **4** may have a simpler and more reliable structure than other backflow preventing means previously proposed by the inventor, such as the backflow preventing means included with the known wearable urinary collection apparatus **101** and (e.g., the entrance port **102**) shown in present FIGS. **1A-1B**.

[0072] For example, referring to FIGS. **7A-7C**, a leakage inhibiting entrance opening (e.g., the expandable entrance opening **6**) may be easily formed at the one end portion of the urine collection receptacle **2** by, for example, inserting a thin, planar and somewhat rigid member **20** within one end portion of the urine collection receptacle **2**, providing a small elastic band **22**, e.g., a rubber band having a diameter of about 2 cm, around the portion of the urine collection receptacle **2** having the thin, planar and somewhat rigid member **20** therein so that the small elastic band **22** is near but spaced inward of one end of the urine collection receptacle **2**, and then folding the one end of the urine collection receptacle **2** back over the small elastic band **22** and removing the thin, planar and somewhat rigid member **20** such that the small elastic band **22** is contained by the folded end portion of the urine collection receptacle **2** and is situated at a new end of the urine collection receptacle **2** to define the expandable entrance opening **6**, which may be referred to as an expandable entrance opening. In this manner the small elastic band **22** and the fold of the plastic sheet material forming the urine collection receptacle **2** in which the small elastic band **22** is contained may become constricted, though easily expandable such that a user may directly insert his penis or penis glans therein. The elastic, expandable nature of the expandable entrance opening **6** as provided by the small elastic band **22** and stretchable plastic sheet material is advantageous for engaging the penis or penis glans with a small force sufficient to prevent urine which is being discharged by the penis from leaking out of the expandable entrance opening **6**, but without causing discomfort to the user's penis. If desired, a lubricant such as a hydrogel or the like may also be applied to surface(s) of the user's penis or penis glans before it is inserted into the expandable entrance opening **6** to help prevent undesired seepage of urine and enhanced comfort. Also, while the expandable entrance opening **6** may be directly formed at one end of the urine collection receptacle **2**, in some implementations, it may be formed as a separate tubular member of plastic sheet material which is attached at one open end of the urine collection receptacle **2** using heat or energy pulses.

[0073] Referring back to FIGS. **4-6**, the backflow preventing **4** means may include a constriction **10** formed in the urine collection receptacle **2** which reduces (e.g., significantly reduces) the ID of the urine collection receptacle **2** at the position of the constriction **10** so that urine discharged into the urine collection receptacle **2** through the expandable entrance opening **6** will flow downward through the constriction **10** for being collected in the urine collection receptacle **2**. The backflow preventing means **4** may also include a small tubular member **8**, e.g., 6.0-9.0 cm long and 1.5-3.0

cm wide, having open opposite ends and formed of plastic sheet material, which has one of its open ends secured to the urine collection receptacle **2** at the constriction **10** such that urine which passes through the constriction **10** will also flow through the small tubular member **8**. Finally, the backflow preventing means **4** may include a check valve **14** also formed of plastic sheet material provided at the one end of the tubular member **8** which is secured to the urine collection receptacle **2** at the constriction **10**. The check valve **14** permits urine to freely pass further inward of the urine collection receptacle **2** through the small tubular member **8**, but blocks the urine from passing back up through the small tubular member **8** and the constriction **10**.

[0074] As described, the backflow preventing means **4** may include the constriction **10** formed widthwise in the urine collection receptacle **2** which significantly reduces the ID of the urine collection receptacle **2** at the position of the constriction **10** to an opening (e.g., the constricted opening **12**) having an ID which may be $\frac{1}{3}$ to $\frac{1}{5}$ of the ID of the urine collection receptacle **2** where the urine collection receptacle **2** is free from the constriction **10**. Stated differently, the constriction **10** may define the constricted opening **12** extending therethrough. Accordingly, urine discharged in the urine collection receptacle **2** through the expandable entrance opening **6** will flow downward through the constriction **10** before being collected in the urine collection receptacle **2**. The small tubular member **8** has open opposite ends, one of which may be secured to the urine collection receptacle **2** at the constriction **10** such that urine which passes through the constricted opening **12** will also flow through the small tubular member **8**. The check valve **14**, also formed of plastic sheet material, provided at the one end of the tubular member **8** is secured to the urine collection receptacle **2** at the constriction **10**. The check valve **14** permits urine to freely pass further inward of the urine collection receptacle **2** through the small tubular member **8**, but blocks the urine from passing back up through the small tubular member **8** and the constriction **10**.

[0075] The small tubular member **8** may be readily formed of plastic sheet material which is fixed into a tubular shape using heat of energy pulses similarly to the urine collection receptacle **2**. While the ID of the small tubular member **8** may be 1.5-3.0 cm, the inventor has found that the bore or lumen of the small tubular member **8** is preferably ≤ 2.0 cm particularly at the end connected at the constriction **10**. The plastic sheet material may be the same plastic sheet material used for forming the urine collection receptacle **2** but may be formed from a different plastic sheet material and/or may have a different thickness than the material used to form the urine collection receptacle **2**. Again, TPU sheet material is preferred for forming each of these components because of its advantageous characteristics discussed herein. However, even if made of the same plastic material as that of the urine collection receptacle **2**, the small tubular member **8** may have a smaller wall thickness than the urine collection receptacle **2** because it is not required to contain any urine therein, unlike the urine collection receptacle **2**, and will remain in a substantially flat or collapsed state at all times, including when urine discharged by a user is passing therethrough. For example, the urine collection receptacle **2** may have a wall thickness of 0.025-0.150 mm (1-6 mil), while the small tubular member **8** and the check valve **14** may have a wall thickness of 0.025-0.075 mm (1-3 mil).

[0076] As described, the small tubular member **8** may be readily formed of plastic sheet material which is fixed into a tubular shape using heat of energy pulses similarly to the urine collection receptacle **2**. While the ID of the small tubular member **8** may be 1.5-3 cm, the inventor has found that the bore or lumen of the member is preferably ≤ 2.0 cm particularly at the end connected at the constriction **10**. The plastic sheet material may be the same plastic sheet material used for forming the urine collection receptacle **2** but may be a different plastic sheet material and/or may have a different thickness than the material used in forming the urine collection receptacle **2**. Again, TPU sheet material is preferred for forming components of the wearable urinary collection apparatus **1** because of its advantageous characteristics discussed herein. However, even if made of the same plastic material as the urine collection receptacle **2**, the small tubular member **8** may have a smaller wall thickness than the urine collection receptacle **2** because it is not required to contain any urine

therein, unlike the urine collection receptacle **2**, and will remain substantially flat or collapsed state at all times, including when urine discharged by a user is passing therethrough. For example, the urine collection receptacle **2** may have a wall thickness of 0.025-0.150 mm (1-6 mil), while the small tubular member **8** and the check valve **14** may have a wall thickness of 0.025-0.075 mm (1-3 mil).

[0077] Referring to FIGS. **4** and **7A-7C**, the expandable entrance opening **6** may, for example, be easily formed at the one end portion of the collection receptacle **2** in a manner as shown in FIGS. **7A** to **7C**. Initially, the thin, planar and somewhat rigid member **20** may be inserted within one end portion of the urine collection receptacle **2**, but spaced away from the an end opening of the urine collection receptacle **2**, and a small, thin gauge elastic band, such as the small elastic band **22**, e.g., a rubber band having a diameter of about 2 cm and thickness of 0.5-1.5 mm, may be provided around the portion of the urine collection receptacle **2** having the thin, planar and somewhat rigid member **20** therein so that the small elastic band **22** is near but spaced inward of one end of the urine collection receptacle **2** as shown in FIG. **7A**. Then, the one end of the urine collection receptacle **2** may be folded back over the small elastic band **22** as shown in FIGS. **7B-7C**. The thin, planar and somewhat rigid member **20** may then be removed from the urine collection receptacle **2** such that the small elastic band **22** is contained by the folded end portion of the urine collection receptacle **2** and is situated at a new end of the urine collection receptacle **2** to define the expandable entrance opening **6** shown in FIG. **4**. In this manner, the small elastic band **22** and the fold of the plastic sheet material forming the urine collection receptacle **2** in which the small elastic band **22** is contained form the constricted, but easily expandable entrance opening (e.g., the expandable entrance opening **6**) of the urine collection receptacle **2** into which a user may directly insert his penis or penis glans. The elastic, expandable nature of the expandable entrance opening **6** as provided by the small elastic band **22** and stretchable plastic sheet material is advantageous for engaging the penis or penis glans with a small force sufficient to prevent urine which is being discharged by the penis from leaking out of the expandable entrance opening **6**, but without causing discomfort to the user's penis. If desired, a lubricant such as a hydrogel or the like may also be applied to surface(s) of the user's penis or penis glans before it is inserted into the expandable entrance opening **6** to help prevent undesired seepage of urine and enhanced comfort.

[0078] Also, while the expandable entrance opening **6** may be directly formed at one end of the urine collection receptacle **2**, the expandable entrance opening **6** may be formed as a separate tubular member of plastic sheet material which is attached at one open end of the urine collection receptacle **2** using heat or energy pulses. Referring to FIGS. **8-11**, for example, there is shown another example of the wearable urinary collection apparatus **1** according to an implementation of the present disclosure. Referring to FIG. **8**, one open end of the urine collection receptacle **2** is folded back over an intermediate portion of the urine collection receptacle **2** and a first of the backflow preventing means **4**, including the constricted opening **12**, the small tubular member **8** and the check valve **14** is provided at one axially inward position along the urine collection receptacle **2**. Referring to FIG. **9**, a part of the one end of the urine collection receptacle **2** that had been folded over is moved back toward its original position and a second of the backflow preventing means **4**, including another of the constricted opening **12**, another of the small tubular member **8**, and the check valve **14** is provided at another position along the urine collection receptacle **2** which is less axially inward than the position of the first of the backflow preventing means **4**. At this time, the remaining folded portion of the one end of the urine collection receptacle **2** extends axially away from the second of the backflow preventing means **4** and defines a new open end of the urine collection receptacle **2**. The new open end may extend out slightly in a lateral direction from the rest of the urine collection receptacle **2**, and this is shown in a somewhat exaggeratedly large shape in FIG. **9**. Such new end of the urine collection receptacle **2**, including folded layers of the plastic sheet material, has enhanced strength compared to other portions of the urine collection receptacle **2** which include only a single layer of the plastic sheet material, and

may be useful for supporting the urine collection receptacle **2** on a user's body through a harness, such as the harness **40** shown in FIG. **12** and/or a supporting sleeve, such as the supporting sleeve **50** shown in FIG. **13**, as discussed further herein.

[0079] In FIG. **10** an example of the expandable entrance opening **6'** that is not formed directly into the one end of the urine collection receptacle **2**, as is done in the implementation of FIGS. **4** and **7A-7C**, but rather the expandable entrance opening **6'** is initially provided in a separate member **30**, which is formed of plastic sheet material such as used in forming the urine collection receptacle **2**, and then the separate member **30** is attached to the urine collection receptacle **2** at its one open end using heat, energy pulses, adhesive or other appropriate means.

[0080] The check valve **14** (e.g., forming part of the backflow preventing means **4**) is a particularly important aspect of the present disclosure. As best shown in FIGS. **5A-5B**, the check valve **14** may be provided in the form of two, separate flaps of plastic sheet material which may each have one end secured at the constriction **10** formed in the urine collection receptacle **2** which significantly reduces the ID of the urine collection receptacle **2** leading into the small tubular member **8** such that the flaps freely extend into the one connected end of the small tubular member **8**, with the flaps being respectively disposed on opposite sides of a constricted opening **12** defined by the constriction **10**. The flaps function as the check valve **14** because when a user discharges urine into the urine collection receptacle **2** (e.g., via the expandable entrance opening **6**), the urine will readily pass through the constricted opening **12** and between the flaps into the small tubular member **8**, but in the event of urine backflowing up through the small tubular member **8** for any reason, the flaps will tend to fold over the constricted opening **12** to block the backflowing urine from passing back out of the small tubular member **8** and through the constricted opening **12**.

[0081] As described, the check valve **14** forming part of the backflow preventing means **4** is a particularly important aspect of the present disclosure. As shown in FIG. **5A**, the check valve **14** may be provided in the form of two, separate flaps of plastic sheet material which may each have one end secured at the constriction **10** formed in the urine collection receptacle **2** such that the two flaps freely extend into the one connected end of the small tubular member **8** from the constriction **10**, with the two flaps being respectively disposed on opposite sides of the constricted opening **12**. The flaps function as the check valve **14** because when a user discharges urine into the expandable entrance opening **6**, the urine will readily pass downward through the constricted opening **12** and between the two flaps into the small tubular member **8**, but in the event of urine backflowing up through the small tubular member **8** for any reason, both flaps will tend to fold over the constricted opening **12** to block the backflowing urine from passing back out of the small tubular member **8** and through the constricted opening **12**.

[0082] Referring to FIG. **5B**, another implementation of the small tubular member **8** is shown. The small tubular member **8** shown in FIG. **5B** is similar to the small tubular member shown in FIG. **5A**. However, in the implementation shown in FIG. **5B**, the attachment position of the end portion of the small tubular member **8** at the constriction **10** is different from the attachment position of the end portion of the small tubular member **8** shown in FIG. **5A** such that the two flaps forming the check valve **14** are fully disposed within the one end of the small tubular member **8** and spaced away from the constriction **10** by a distance **d**. In other words, ends of the slits cut into the one end of the small tubular member **8** to define the flaps of the check valve **14** end at the position **13** and are spaced away from the constriction by the distance **d**. With the check valve **14** shown in FIG. **5B**, a portion of the one end of the small tubular member **8** extends inward of the small tubular member **8** by the distance **d**, while the flaps of the check valve **14** extend further inward of the small tubular member **8** from the position **13**. As with the implementation shown in FIG. **5A**, each of the flaps of the check valve **14** should have sufficient axial length to fully cover the constricted opening **12**, while one of the flaps may have a greater axial length than that of the other flap.

[0083] The flaps may be formed of the same plastic sheet material as used in forming the small tubular member **8** and may be conveniently and inexpensively formed integrally with the small

tubular member **8**. For example, the flaps may initially be provided as one end of the small tubular member **8**, and two opposing slits may be formed in the one end portion of the tubular member **8** so as to define the flaps. Then, the flaps may be folded inward of the one end of the tubular member **8**, and when the one end of the small tubular member **8** is attached at the constriction **10**, the flaps may also be secured at different parts of the constriction **10**. The flaps may be rectangular, square or some other shape, and may have different lengths, but each of the flaps should be sufficiently long such that when it is folded in the direction of the constricted opening **12**, it may fully cover the constricted opening **12**. For example, if the constricted opening **12** of the urine collection receptacle **2** at which the small tubular member **8** is attached is 1 cm in diameter and each of the flaps is connected to the constriction **10** 0.5 cm away from the constricted opening **12**, each of the flaps should have a length of at least 1.6 cm. If the flaps have different lengths, e.g., one is 0.2-0.5 cm longer than the other, in the event of urine backflowing up through the small tubular opening **8**, the shorter of the flaps will tend to fold over the constricted opening **12** first and the longer of the flaps will then tend to fold over the shorter of the flaps.

[0084] As described, the flaps of the check valve **14** may be formed of the same plastic sheet material as used in forming the small tubular member **8**, and may be conveniently and inexpensively formed integrally with the small tubular member. Referring to FIG. **6**, for example, the two flaps may initially be provided as one end of the small tubular member **8** by forming two opposing slits in the one end portion of the tubular member **8** in an axial direction of the small tubular member **8** so as to define the flaps. Thereafter, the two flaps may be folded inward of the one end of the small tubular member **8**, and when the one end of the small tubular member **8** is attached at the constricted opening **12**, the two flaps may also be secured at different parts of the constriction **10** on opposite sides of the constricted opening **12**. Each of the two flaps may have a curved shape corresponding to about half of the small tubular member **8**, but the flaps may be rectangular, square or some other shape, and may have different axial lengths so that they extend inward of the small tubular member **8** by different the different lengths. However, each flap should be sufficiently long such that when it is folded in the direction of the constricted opening **12** it may fully cover the constricted opening **12**. For example, if the constricted opening **12** is 1 cm in diameter and each of the flaps is connected at the constriction **10** about 0.5 cm away from the constricted opening **12**, each flap should have a length of at least 1.6 cm. If the two flaps have different lengths, e.g., one is 0.2 to 0.5 cm longer than the other, in the event of urine backflowing up through the small tubular member **8**, the shorter flap will tend to fold over the constricted opening **12** first and the longer flap will then tend to fold over the shorter flap. The rounded edges of the constricted opening **12** where the small tubular member is connected at the constriction **10** help to prevent the constricted opening **12** from closing when a user has discharged urine into the urine collection receptacle **2** and the urine flows inward of the urine collection receptacle **2** through the constricted opening **12** before being collected in the urine collection receptacle **2**.

[0085] While backflow preventing means according to the present disclosure may comprise a single constriction **10** and single tubular member **8** having the check valve **14** at one end, as shown in FIG. **4**, it is possible to provide a plurality of the constrictions **10** formed in spaced positions from each other along the urine collection receptacle **2**, and a plurality of the small tubular members **8** and check valves **14** provided with the constrictions **10**, respectively. For example, each of the constrictions **10** may be provided at the lateral center of the urine collection receptacle **2** and spaced from each other by 7-10 cm along the length of the urine collection receptacle **2**, with one of the small tubular members **8** and one of the check valves **14** provided at each of the constrictions **10**. The inventor has determined through experimentation that two of the constrictions **10**, each having one of the small tubular members **8** and one of the check valves **14** provided therewith, is very effective at preventing any backflow, even when the urine collection receptacle **2** contains a relatively large amount of urine, e.g. >8 fluids ounces.

[0086] As described, while the wearable urinary collection apparatus **1** according to the present

disclosure may include a single one of the backflow preventing means **4** comprising the constriction **10**, the small tubular member **8**, and the check valve **14**, it is possible to provide a plurality of the backflow preventing means **4** provided in spaced positions from each other along the urine collection receptacle **2**. For example, each of the backflow preventing means **4** may be provided at the lateral center of the urine collection receptacle **2** and spaced from each other by 7-10 cm along the length of the urine collection receptacle **2**. The inventor has determined through experimentation that two of the backflow preventing means **4**—each having one of the constrictions **10**, one of the small tubular members **8**, and one of the check valves **14** provided therewith—is very effective at preventing any backflow, even when the collection receptacle contains a relatively large amount of urine, e.g. 28 fluids ounces.

[0087] Additionally, the inventor has determined that if a manipulation cord or other member is provided at a lower end the urine collection receptacle **2**, such as the cord **109** shown in FIG. **1A**, the cord may be easily used to manipulate the lower portion of the urine collection receptacle **2** relative to the backflow preventing means **4** to further assure positively all backflow of urine out of the urine collection receptacle **2** is prevented. For example, the inventor has determined that if the cord is used to fold the lower portion of the urine collection receptacle **2** relative to an intermediate portion of the urine collection receptacle **2** inward of the backflow preventing means **4**, the combination of the folded lower portion of the urine collection receptacle **2** and the backflow preventing means **4** is very effective for assuring that no backflow will occur.

[0088] As described, the inventor has determined that if a manipulation cord is provided at a lower end the collection receptacle **2**, such as the cord **109** shown in FIG. **1A**, the cord may be used to easily manipulate the lower portion of relative to the backflow preventing means **4** to further assure that positively any backflow of urine in the urine collection receptacle **2** is prevented. For example, the inventor has determined that if the cord **109** is used to fold the lower portion of the urine collection receptacle **2** relative to an intermediate portion of the urine collection receptacle **2** that is positioned inward of the backflow preventing means **4**, the combination of the folded lower portion of the urine collection receptacle **2** and the backflow preventing means **5** is very effective for assuring that no backflow will occur from the urine collection receptacle **2** back up through the backflow preventing means **4**.

[0089] According to a second discovery, the inventor has determined that the harness **104** disclosed in the '535 patent and duplicated in present FIG. **2A** may not always hold the known wearable urinary collection apparatus **101** at a position sufficiently close to a user's pubic area to permit the user to easily discharge urine into the urine collection receptacle **100** at all times. After considerable research, the inventor has developed an improved implementation of a harness **40** for supporting the urine collection receptacle **2** for consistent convenience in use. Referring to FIG. **12**, the harness **40** includes a waist strap **42** that may be secured about a user's waist, a pair of supporting straps **44** that have opposite ends connected to different parts of the waist strap **42** such that portions of the supporting straps **44** extend on opposite sides of the user's pubic area, and a short, e.g., 2-4 cm, bridge connection **46** that connects intermediate portions of the supporting straps **44** at a level of the user's inferior pubic tubercle. The harness **40** defines a substantially triangular area between the waist strap **42**, portions of the supporting straps **44** that extend from the waist strap **42**, and the bridge connection **46**, such that the substantially triangular area closely surrounds the user's pubic area. The inventor has determined that if the urine collection receptacle **2** is secured to the harness **40** such that the expandable entrance opening **6** of the urine collection receptacle **2** is supported by the supporting straps **44** close to the bridge connection **46**, the user may easily discharge urine in the urine collection receptacle **2** at all times.

[0090] As described, the inventor has determined that the harness **104** disclosed in the '535 patent and duplicated in present FIGS. **3A-3B** may not always hold an elongate, tubular collection receptacle, such as the urine collection receptacle **2**, at a position sufficiently close to a user's pubic area to permit the user to easily discharge urine in the urine collection receptacle **2** at all times.

After considerable research, the inventor has conceived of an improved harness for supporting an elongate, tubular collection receptacle such as the urine collection receptacle **2** for consistent convenience in use. Referring to FIG. **4**, the harness **40** includes the waist strap **42** that may be secured about a user's waist, a pair of supporting straps **44** that have opposite ends connected to different parts of the waist strap **42** such that when a user wears the harness **40**, portions of the supporting straps **44** extend on opposite sides of the user's pubic area, and a short, e.g., 2-4 cm, bridge connection (e.g., the bridge connection **46**) that connects intermediate portions of the securing straps **116** at a level of the user's inferior pubic tubercle. The harness **40** defines a substantially triangular area between the waist strap **42**, portions of the supporting straps **44** that extend from the waist strap **42**, and the bridge connection **46**, such that the substantially triangular area closely surrounds the user's pubic area. The inventor has determined that if the urine collection receptacle **2** is secured to the harness **40** such that the expandable entrance opening **6** of the urine collection receptacle **2** is supported by the supporting straps **44** and/or the bridge connection **46**, the user may be enabled to easily discharge urine into the urine collection receptacle **2** at all times. [0091] Again, while the urinary collection apparatus **1** has been discussed as being used by male persons, it may also be used by females, and this typically involves use of an adapter. Appropriate adapters have been previously proposed by the present inventor in the '535 Patent, e.g., see FIGS. **3A-3E** of the '535 patent, and such adapters may be used together with the wearable urinary collection apparatus **1** according to the exemplary implementation of the present disclosure. The inventor has determined that the harness **40** may also be used by females, but for such use it may be desirable to modify the harness **40** by addition of a second short bridge connection (not shown) which is the same or similar to the bridge connection **46**. The second short bridge connection may also be connected to the supporting straps **44** such that it extends in parallel to the bridge connection **46** but spaced away from the bridge connection by about 2-4 cm. With such modified harness including the second short bridge connection, a discharge end of the female adapter may be extended between the two short connections and into the expandable entrance opening **6** of the urine collection receptacle **2**.

[0092] The urine collection receptacle **2** may be directly supported by the harness **40**, e.g., by securing an edge of the urine collection receptacle **2** which surrounds the expandable entrance opening **6** in FIG. **10** to the supporting straps **44** and/or the bridge connection **46** using any appropriate fastening means. The inventor has determined that if the expandable entrance opening **6** is disposed closely adjacent to the supporting straps **44** and/or the bridge connection **46**, this will be effective for assuring convenience in use.

[0093] The inventor has also determined that the known wearable urinary collection apparatus **101** may also include a supporting sleeve **50** formed of fabric or the like to contain the urine collection receptacle **2** for additional support and comfort to the user. Referring to FIGS. **13-14**, the supporting sleeve **50** may have an elongate tubular shape similar to the urine collection receptacle **2** so that the urine collection receptacle **2** fits securely therein without restricting the ability for the urine collection receptacle **2** to easily expand when it collects urine therein. The supporting sleeve **50** may have one open end to which the open end of the urine collection receptacle **2** may be secured, and a zipper **54** or other fastener may be provided with a portion of the supporting sleeve **50** adjacent to the open end to permit the end to be selectively expanded to facilitate insertion and removal of the urine collection receptacle **2** into and from the supporting sleeve **50**. Additionally, the open end of the supporting sleeve **50** may include hooks, snaps, straps or any appropriate fastener **56** which may be used to secure the open end of the urine collection receptacle **2** (e.g., the expandable entrance opening **6**) to the open end of the supporting sleeve **50** and to secure the open end of the supporting sleeve **50** to the harness **40**. The size of the open end of the supporting sleeve **50** may be sufficiently large that a male user may insert his scrotum, as well as his penis or penis glans into the open end of the supporting sleeve **50**. This feature helps to maintain the open end of the urine collection receptacle **2**—as disposed in the supporting sleeve **50**—conveniently close to

the user's penis or penis glans. The open end of the urine collection receptacle **2** may be configured to comfortably receive only the user's penis or penis glans therein. The opposite end of the supporting sleeve **50** may be closed or may include some a closure that may be selectively opened or closed to expose the opposite end of the urine collection receptacle **2** or a portion thereof.

[0094] As described, the inventor has also determined that the wearable urinary collection apparatus **1** may also include a supporting sleeve (e.g., the supporting sleeve **50**) formed of fabric or the like to contain the collection receptacle for additional support and comfort to the user. An exemplary implementation of the supporting sleeve **50** is shown in FIGS. **13-14**. As shown, the supporting sleeve **50** may have an elongate tubular shape similar to the urine collection receptacle **2** so that the urine collection receptacle **2** fits securely therein but without restricting the ability of the urine collection receptacle **2** to easily expand when it collects urine therein. The supporting sleeve **50** may have a first open end **52** to which the open end of the urine collection receptacle **2** may be secured, and a zipper (e.g., the zipper **54**) or other fastener may be provided with a portion of the supporting sleeve **50** adjacent to the first open end **52** to permit the first open end **52** to be selectively expanded to facilitate insertion and removal of the urine collection receptacle **2** into and from the supporting sleeve **50**. Also, the first open end **52** of the supporting sleeve **50** may include hooks, snaps, straps, hook-and-loop fasteners or the like (e.g., the fastener **56**) which may be used to secure the open end of the urine collection receptacle **2** to the first open end **52** of the supporting sleeve **50** and to secure the first open end **52** of the supporting sleeve **50** to the harness **40**. The size of the first open end **52** of the supporting sleeve **50** may be sufficiently large that a male user may insert his scrotum, as well as his penis or penis glans, into the first open end **52** of the supporting sleeve **50**. This may help to maintain the first open end **52** of the urine collection receptacle **2**, as disposed in the supporting sleeve **50**, conveniently close to the user's penis or penis glans therein. As shown in FIG. **14**, for example, the expandable open end **6** of the urine collection receptacle **2** may be secured non-coaxially within the first open end **52** of the supporting sleeve **50** such that part of the first open end **52** does not have any part of the expandable open end **6** disposed therein and remains open so that the user may readily insert his scrotum into the supporting sleeve **50** through such part of the first open end **52**. The inventor has determined that inserting and maintaining one's scrotum into the first open end **52** of the supporting sleeve **50** in this manner does not create any discomfort for the user. An opposite end **58** of the supporting sleeve **50** may be open as depicted to permit easy access to the lower end of the urine collection receptacle **2**. Alternatively, the opposite end **58** of the supporting sleeve **50** may be closed or may include some type of closure that may be selectively opened or closed to expose the opposite end **58** of the urine collection receptacle **2** or a portion thereof.

[0095] According to a third discovery by the present inventor, the known elastomeric cap **110** as shown in FIGS. **3A-3B** may not be sufficiently comfortable for a user to have provided around his penis or penis glans for an extended period of time. After doing further research, the inventor has developed a more appropriate leakage preventing cap for being provided around his penis or penis glans. Referring to FIG. **4**, the leakage preventing cap **60** may be formed of a highly elastic, and tensile silicone gel material which is formed into a short tubular shape with opposite open ends. The leakage preventing cap **60** may be 2-5 cm long, have an inner diameter of 1.5-3.0 cm, and have a wall thickness of 1-3 mm. With such construction, the leakage preventing cap **60** may be easily and comfortably fitted over most user's penis or penis glans such that the leakage preventing cap **60** snugly engages the penis or penis glans, as well as the expandable entrance opening **6** of the urine collection receptacle **2** to prevent urine from leaking out of the expandable entrance opening **6**, but will not create any discomfort for the user even if the leakage preventing cap **60** remains on the user's penis or penis glans for an extended period time (e.g., an hour or more).

[0096] As described the inventor has also determined that the wearable urinary collection apparatus **1** may also include more reliable and comfortable leakage preventing cap (e.g., the leakage preventing cap **60**) for being provided around a user's penis or penis glans prior to the penis or

penis glans being inserted into the expandable entrance opening **6**, such as shown in FIG. **5**. The leakage preventing cap **60** is a separate member which is simply fitted around the user's penis or penis glans but not connected to the urine collection receptacle **2** or any other part of the wearable urinary collection apparatus **1**. The leakage preventing cap **60** may be formed of a highly elastic, and tensile silicone gel material which is formed into a short tubular shape with opposite open ends. The leakage preventing cap **60** may be 2-5 cm long, have an inner diameter of 1.5 to 3 cm, and have a wall thickness of 1-3 mm. With such construction the leakage preventing cap **60** may be easily and comfortably fitted over most user's penis or penis glans such that the leakage preventing cap **60** snugly engages the penis or penis glans, as well as the expandable entrance opening **6** of the urine collection receptacle **2**, to prevent urine from leaking out of the entrance opening **6**, but will not create any discomfort for the user even if the leakage preventing cap **60** remains on the user's penis or penis glans for an extended period (e.g., an hour or more).

[0097] As described previously, FIG. **10** shows an example of the expandable entrance opening **6'** that is not formed directly into the one end of the urine collection receptacle **2**, as is done in the implementation of FIGS. **4** and **7A-7C**, but rather the expandable entrance opening **6'** is initially provided in a separate member **30**, which is formed of plastic sheet material such as used in forming the urine collection receptacle **2**, and then the separate member **30** is attached to the urine collection receptacle **2** at its one open end using heat, energy pulses, adhesive or other appropriate means. The separate member **30** having the expandable entrance opening **6'** may be formed similarly to how the expandable entrance opening **6** is formed with reference to FIGS. **7A-7C** using the small elastic band **22**, the planar, somewhat rigid member **20**, and a short tubular member formed of plastic sheet material having an ID corresponding to that of the urine collection receptacle **2**, e.g., the planar, somewhat rigid member **20** may be inserted into the tubular member, the small elastic band **22** may be placed around the tubular member where the planar, somewhat rigid member **20** is located, the tubular member may be folded itself from one end toward the other, and then the planar, somewhat rigid member **20** may be removed, thereby defining the expandable entrance opening **6'** at one end of the folded tubular member. When the separate member **30** is connected to the open end of the urine collection receptacle **2**, a lower open end of the separate member **30** opposite to the expandable entrance opening **6'**, may be connected to the open upper end of the urine collection receptacle **2** slightly inward of the edge of open end where the broken line extends in FIG. **10** using heat, energy pulses or the like and such that the lower end of the folded tubular member is spaced away from the closest one of the backflow preventing means **4**.

[0098] The connection between the separate member **30** and the urine collection receptacle **2** may not create any constriction to the ID of the collection receptacle such as in FIG. **10**, but it is possible to create a constriction at such connection. For example, FIG. **11** shows a modification to structure of FIG. **10** wherein the separate member **30** is connected to the urine collection receptacle **2** at an axially intermediate part of the separate member via a constriction **32** which reduces the ID of the collection receptacle to about $\frac{1}{3}$ to $\frac{1}{5}$ of its original size, and such that a portion of the separate member **30** apposite to the expandable opening **6'** extends inward of the constriction **32**. With this modification, the constriction **32** and the lower portion of the separate member **30** may act as a type of backflow preventing means with or without one or more of the backflow preventing means **4** disposed inward of the separate member **30** within the urine collection receptacle **2**.

[0099] FIG. **15** is an upper perspective view illustration of an example of a wearable urinary collection apparatus **200** according to the present disclosure. The wearable urinary collection apparatus **200** comprises a urine collection receptacle **210** and an inner collection member **212** that are cooperatively configured to efficiently collect and store urine while ensuring user comfort and hygiene. As described in greater detail below, the inner collection member **212** may be disposed within a redundant urine collection area of the urine collection receptacle **210** such that a stream guide tube extending from an end of the inner collection member **212** extends into the urine collection receptacle **210** to form a fluid-flow path from the inner collection member **212** directly

into the urine collection receptacle **210**. Through careful study of existing technologies, such as the known wearable urinary collection apparatus **101** and the wearable urinary collection apparatus **1**, the inventor has developed certain improvements to address critical aspects such as user comfort, backflow prevention, leak prevention, manufacturability, and reusability.

[0100] The wearable urinary collection apparatus **200** represents a significant advancement over previous implementations, incorporating innovative features to enhance its overall performance and user experience. The urine collection receptacle **210** and the inner collection member **212** are configured with a focus on efficient urine storage and leak prevention, and may be constructed from flexible, durable materials that may comfortably conform to the user's body while maintaining structural integrity.

[0101] As described in further detail below, an improvement of the wearable urinary collection apparatus **200** is the incorporation of one or more mechanisms, such as one-way valves or strategically placed constrictions, to ensure that urine flows only in the intended direction—from the user into the urine collection receptacle **210**. Furthermore, manufacturability has been carefully considered in the wearable urinary collection apparatus **200**. The components thereof, as described in further detail below, may be configured for ease of assembly and cost-effective production. Finally, reusability is another important aspect addressed in this improved implementation. While some components may be disposable for hygiene reasons, the overall structure of the wearable urinary collection apparatus **200** may be configured to be durable and easy to clean. This approach not only reduces waste but also makes the device more economical for long-term use.

[0102] The wearable urinary collection apparatus **200** may be configured for both ambulatory and stationary applications. This flexibility ensures that the device is suitable for a wide range of users, including those with temporary or permanent mobility issues, as well as individuals in various occupational settings where access to traditional restroom facilities may be limited.

[0103] FIG. **15** is an upper perspective view illustration of the wearable urinary collection apparatus **200**. The inner collection member **212** may be configured to receive at least a portion of a length of a penis **202** of a user. The inner collection member **212** may extend longitudinally from an upper end to a lower end. The inner collection member **212** may include a penile sleeve **230** configured to extend along a shaft **204** of the penis **202**. The penile sleeve **230** may extend from a voiding area **232** (e.g., an end portion of the inner collection member **212**) toward the upper end of the inner collection member **212**. The voiding area **232** may be disposed at the lower end of the inner collection member **212** and may be configured to be in close proximity (e.g., adjacent) to a penis glans **206** and/or a urethral opening **208** (e.g., urethra) of the penis **202**. In other words, the voiding area **232** (e.g., an end portion of the inner collection member **212**) may be configured to extend around the penis glans **206** of the penis **202**. This strategic placement of the voiding area **232** allows for efficient capture of the urinary fluid **214** as it is voided, directing it immediately into the urine collection receptacle **210** and minimizing the risk of leakage or backflow.

[0104] The inner collection member **212** may include a stream guide tube **234** extending from the voiding area **232** (e.g., away from the upper end). In some implementations, the voiding area **232** may be configured to be adjacent to a penis glans **206** of the penis **202**. In some implementations, the stream guide tube **234** may be configured to be adjacent to a terminal end of the penis **202**, such as the urethral opening **208** (e.g., urethra). The stream guide tube **234** may be formed from or comprise a flexible elastomer and/or may be relatively rigid. In some implementations, the stream guide tube **234** may be about 5 cm long and may have an inside diameter of about 1.0 cm. The stream guide tube **234** is configured to efficiently direct the urinary fluid **214** from the voiding area **232** into the urine collection receptacle **210** by ensuring that urine is guided directly into the urine collection receptacle **210** without risk of leakage or backflow. One or both of the voiding area **232** and the stream guide tube **234** may be formed from or comprise a silicon membrane, an elastomer, or a combination thereof.

[0105] In some implementations, the stream guide tube **234** may include a one-way valve **236**

disposed at a terminal end thereof. The one-way valve **236** may be or comprise a thin (e.g., 1-2 mil) polymer tube that is about 2-4 cm long and flexible. The one-way valve **236** may be formed from or comprise the same material as that of the urine collection receptacle **210**. As shown, in some implementations, the one-way valve **236** may comprise a flexible sheet adjacent to an opposing surface. The flexible sheet may be biased toward the opposing surface when the urinary fluid **214** is not flowing through the stream guide tube **234** to inhibit the urinary fluid **214** from traveling through the stream guide tube **234** from within the urine collection receptacle **210**.

[0106] When the urinary fluid **214** flows through the stream guide tube **234**, the flexible sheet may be urged away from the opposing surface to enable the urinary fluid **214** to travel through the stream guide tube **234** from the inner collection member **212**. In other words, when the user voids themselves of the urinary fluid **214**, the pressure of the flowing fluid pushes the flexible sheet away from the opposing surface, allowing the urine to pass therethrough into the urine collection receptacle **210**. As explained in further detail below, the one-way valve **236** may be inserted through a contractile stem tube of the urine collection receptacle **210** during assembly, which may form a constriction along the stream guide tube **234**. Alternatively, as also explained in further detail below, the one-way valve **236** may be connected to the terminal end of a tubular port of a redundant urine collection area of the urine collection receptacle **210** (described below).

[0107] The material composition and structural characteristics of the stream guide tube **234** and the voiding area **232** are carefully considered to balance flexibility and rigidity. For example, the relative rigidity of the stream guide tube **234** may be important for maintaining its shape and ensuring a consistent urine flow direction, even under varying pressures or user movements. The 5 cm length of the stream guide tube **234** may provide sufficient distance to guide the urinary fluid **214** from the penis **202** into the urine collection receptacle **210**, while the 1.0 cm inside diameter of the stream guide tube **234** may allow for adequate flow capacity without being overly large or cumbersome. An upper band **228** may be included near the upper end of the inner collection member **212** (e.g., opposite the voiding area **232**). In some implementations, the upper band **228** may help secure the inner collection member **212** to the penis **202** (e.g., may aid in rolling the inner collection member **212** onto the penis **202**).

[0108] FIG. **16A** is an upper perspective view illustration of the urine collection receptacle **210** prior to formation of a redundant urine collection area (described below). The urine collection receptacle **210** may be configured to collect urinary fluid **214** therein. As best shown in FIG. **16A**, in some implementations, the urine collection receptacle **210** may comprise or be formed from a tube of flexible polymer film (e.g., membrane) with a thickness of about 3 mil (e.g., about 0.075 cm). The urine collection receptacle **210** may have a flat lying width of about 11 cm and a length (measured orthogonal to the width) of about 80 cm. Accordingly, in some implementations, the urine collection receptacle **210** may be referred to as a membrane (e.g., a tubular membrane, etc.). The urine collection receptacle **210** may extend longitudinally from an upper end to a lower end. As shown in FIG. **16A**, the urine collection receptacle **210** may have an elongated tubular structure with varying diameters (e.g., d, d', d'', etc.) along its length.

[0109] The urine collection receptacle **210**, the inner collection member **212**, and/or components thereof may comprise or be formed from a thin membrane that is generally impermeable to and resistant to the urinary fluid **214**. For example, in some implementations, the urine collection receptacle **210**, the inner collection member **212**, and/or components thereof may comprise or be formed from a thermoplastic polyurethane (TPU) membrane. The use of thermoplastic polyurethane may provide benefits such as durability, flexibility, and ease of cleaning, which may contribute to the overall reusability of the wearable urinary collection apparatus **200**. For example, the durability of TPU may ensure that the urine collection receptacle **210** and/or the inner collection member **212** are capable of withstanding repeated use and the stresses associated therewith, which may reduce the need for frequent replacements. The flexibility of TPU may also allow the urine collection receptacle **210** and/or the inner collection member **212** to conform to the

user's body movements, enhancing comfort and reducing the risk of leaks. Additionally, TPU's smooth surface and chemical resistance make it easy to clean and disinfect, which is crucial for maintaining hygiene in a reusable urinary collection apparatus, such as the wearable urinary collection apparatus **200**.

[0110] Referring to FIGS. **15-16C**, the urine collection receptacle **210** may include the redundant urine collection area **216** (e.g., a redundant fluid collection portion) at the upper end configured to receive the inner collection member **212**. In other words, the urine collection receptacle **210** may include the redundant urine collection area **216** that is configured to extend around the inner collection member **212** or a portion (e.g., a length) thereof.

[0111] FIG. **16B** is an upper perspective view illustration of the urine collection receptacle **210** demonstrating a method for forming the redundant urine collection area **216** according to an example. As shown, the redundant urine collection area **216** may be formed by inverting the upper end of the urine collection receptacle **210** (as shown in FIG. **16B**) inwards at an inversion **218** (depicted in FIG. **16B** by bolded arrows) such that once inverted, the urine collection receptacle **210** extends upwards from the lower end and terminates at the inversion **218**. Furthermore, as explained below with reference to FIGS. **18-20**, in some implementations, the redundant urine collection area **216** may be formed by inverting the upper end of the urine collection receptacle **210** inwards over a rounded rim (e.g., an inner collection ring described below). In the illustrated implementation it may appear as though the inversion **218** is a rigid structure; however, as explained previously, due to the urine collection receptacle **210** being formed from a thin membrane (e.g., a TPU membrane), the inversion **218** may be substantially compliant and flexible, such that, for example, the upper end of the redundant urine collection area **216** may be collapsable under clothes (e.g., a shirt) of the user.

[0112] The redundant urine collection area **216** may extend into the urine collection receptacle **210** to a depth of about 20 cm or longer (e.g., such that a distance between the inversion **218** and an apex of the redundant urine collection area **216** is about 20 cm or longer). The redundant urine collection area **216** serves multiple functions. For example, by inverting the upper end of the urine collection receptacle **210** inwards at the inversion **218**, a double-walled section is formed at the top of the urine collection receptacle **210**. This inverted section forms a pocket-like area that may accommodate the inner collection member **212**, allowing for a secure connection between the two main components of the wearable urinary collection apparatus **200** at an interface **276**.

[0113] The depth of approximately 20 cm or longer for the redundant urine collection area **216** (e.g., the apex of the redundant urine collection area **216**) may be significant for several reasons. Firstly, this depth provides ample space to envelop the inner collection member **212** (e.g., to envelope all of the inner collection member **212** or a portion, such as a length, thereof). This depth also creates additional volume within the urine collection receptacle **210**, effectively increasing its overall capacity for collection of the urinary fluid **214**. Moreover, the redundant urine collection area **216** may act as a safeguard against potential leaks. For example, if any of the urinary fluid **214** were to escape past the primary seal formed by the interface **276** between the urine collection receptacle **210** and the inner collection member **212** (e.g., a stream guide tube **234** and a contractile stem tube **226** described below), it would be contained within this secondary collection area. Finally, due to the flexible nature of the redundant urine collection area **216** (e.g., formed from thin polymer material), the redundant urine collection area **216** may be configured to allow movement of the urine collection receptacle **210** relative to the user by deforming to enable the inner collection member **212** to move relative to the urine collection receptacle **210** without compromising the seal formed between the inner collection member **212** and the urine collection receptacle **210** at the interface **276**.

[0114] The urine collection receptacle **210** may include a tubular port **224** disposed within the redundant urine collection area **216** (e.g., at an apex of the redundant urine collection area **216**) configured to receive the stream guide tube **234** of the inner collection member **212**. In some

implementations, the tubular port **224** may be formed from or comprise thin contractile rubber. In some implementations, a contractile stem tube **226** may be disposed within the tubular port **224**. Where the urine collection receptacle **210** includes the contractile stem tube **226**, one or both of the contractile stem tube **226** and the tubular port **224** may be formed from or comprise thin contractile rubber. The contractile stem tube **226** and/or the tubular port **224** may extend into the urine collection receptacle **210** when the upper end thereof is inverted to form the redundant urine collection area **216**. The combination of the stream guide tube **234** and the contractile stem tube **226** and/or the tubular port **224** may provide a secure connection between the inner collection member **212** and the urine collection receptacle **210** at the interface **276**. As explained previously, the interface **276** may be configured to prevent leaks (e.g., from within the urine collection receptacle **210** into the redundant urine collection area **216**).

[0115] FIG. **16C** is an upper perspective view illustration of the urine collection receptacle **210** demonstrating a method for inserting the inner collection member **212** into the redundant urine collection area **216** according to an example. As shown, the inner collection member **212**—initially positioned above the urine collection receptacle **210**—may be moved (e.g., translated) downward toward the urine collection receptacle **210** and into the redundant urine collection area **216**. When the inner collection member **212** is received in the redundant urine collection area **216**, the stream guide tube **234** may extend through the contractile stem tube **226** and/or the tubular port **224** into the urine collection receptacle **210** to define a fluid flow path from the voiding area **232** to the urine collection receptacle **210**. As described previously, the one-way valve **236** may be disposed at an end (e.g., a terminal end) of the stream guide tube **234** opposite the voiding area **232** and may be configured to inhibit the urinary fluid **214** from backflowing from the urine collection receptacle **210** through the stream guide tube **234**.

[0116] The stream guide tube **234** may also be configured to ensure that the urinary fluid **214** is guided directly into the urine collection receptacle **210** without risk of leakage or backflow. When the inner collection member **212** is properly inserted into the redundant urine collection area **216**, the stream guide tube **234** may pass through the contractile stem tube **226** such that the terminal end thereof (e.g., the one-way valve **236**) is disposed within the urine collection receptacle **210**, creating a secure pathway for the urinary fluid **214** to flow from the user directly into the urine collection receptacle **210**. The interface **276** between the contractile stem tube **226** and the stream guide tube **234** may be configured to create a seal that inhibits the movement of collected urine from the main body of the urine collection receptacle **210** into the redundant urine collection area **216**. This cooperative sealing action may be important for maintaining the efficiency of the system and preventing unnecessary filling of the redundant area.

[0117] The tubular port **224** of the urine collection receptacle **210** may be formed from or comprise the same material as a remainder of the urine collection receptacle **210** (e.g., may be formed from or comprise a TPU membrane). In some implementations, however, the tubular port **224** may be formed from or comprise a different material than that of the remainder of the urine collection receptacle **210**. For example, the tubular port **224** may be formed from or comprise a tubular elastomer that is connected to or formed integrally with the urine collection receptacle **210** (e.g., at an apex of the redundant urine collection area **216**). In some implementations, an inner diameter of the tubular port **224** (e.g., a relaxed inner diameter of the tubular port **224**) may be smaller than an outer diameter of the contractile stem tube **226** (e.g., a relaxed outer diameter of the tubular port **224**) such that the contractile stem tube **226** is securable within the tubular port **224** (e.g., by a press-fit engagement).

[0118] Furthermore, the contractile nature of the tubular port **224** may enable the tubular port **224** to grip the contractile stem tube **226**, forming a seal therebetween that inhibits leaks at the interface between these components (e.g., from within the urine collection receptacle **210** into the redundant urine collection area **216**). The combination of the tubular port **224** and the contractile stem tube **226** may form a multi-layered barrier against leaks and backflow. The contractile properties of the

tubular port **224** and/or the contractile stem tube **226** may also allow passive adjustment to movement or changes in pressure, maintaining their sealing effectiveness during user activity. [0119] The contractile stem tube **226** may comprise a tubular elastomer that has a relaxed inner diameter that is smaller than an outer diameter of the stream guide tube **234**. Accordingly, in some implementations, the contractile stem tube **226** may also be operable to form a constriction along the stream guide tube **234** when the stream guide tube **234** is inserted therethrough. In implementations where the tubular port **224** comprises a tubular elastomer, the stream guide tube **234** and the tubular port **224** may cooperatively form the constriction along the stream guide tube **234**. The constriction formed along the stream guide tube **234** may be configured to enable only unidirectional flow of the urinary fluid **214** from the inner collection member **212** to the urine collection receptacle **210**. Stated differently, the constriction formed along the stream guide tube **234** may enable the urinary fluid **214** to flow from the inner collection member **212** to the urine collection receptacle **210**, and may inhibit (e.g., prevent) the urinary fluid **214** from flowing from the urine collection receptacle **210** to the inner collection member **212**. For example, as the inner collection member **212** is inserted into the redundant urine collection area **216**, the stream guide tube **234** may pass through the contractile stem tube **226** and/or the tubular port **224**. As the urinary fluid **214** flows through the stream guide tube **234**, the contractile stem tube **226** constricts therearound, creating a dynamic seal that adapts to changes in pressure and flow. This constriction helps prevent urine from flowing back up the stream guide tube **234** towards the user, even if the urine collection receptacle **210** is subjected to external pressure or movement.

[0120] In some implementations, select surfaces of the wearable urinary collection apparatus **200** may be coated with hydrogel to provide additional protection against backflow and leakage. For example, the inner surface of the inversion **218**, the upper portion of the redundant urine collection area **216**, and/or the upper band **228** of the inner collection member **212** may be coated with hydrogel to further inhibit leakage of the urinary fluid **214** from the inner collection member **212** and/or the urine collection receptacle **210**. Furthermore, in some implementations, an adaptor (not shown) comprising a female pad with a duct may be connected to the tubular port **224** for use by females. This adaptor may allow the wearable urinary collection apparatus **200** to be used by individuals without a penis. By incorporating both active (constriction) and passive (redundant collection area) measures, the configuration aims to provide reliable performance under various conditions and user movements. This comprehensive approach to fluid management may be important for ensuring user comfort, maintaining hygiene, and preventing potential embarrassment or discomfort that could arise from leaks or backflow.

[0121] Still referring to FIGS. **15-16C**, the urine collection receptacle **210** may include elongate mating members **222** that extend generally away therefrom (e.g., generally away from an upper portion of the urine collection receptacle **210**). In some implementations, the elongate mating members **222** may be connected to an outer surface of the urine collection receptacle **210** using an adhesive or the like. The elongate mating members **222** may include mating features **246** that are removably couplable to complementary mating features of a harness (described below). This configuration may allow for easy attachment and detachment of the urine collection receptacle **210** to and from the harness, facilitating cleaning and replacement when necessary. Although the mating features **246** are illustrated as apertures that extend through the elongate mating members **222**, any suitable type of coupling or geometry may be used (e.g., snaps, magnets, hooks, etc.).

[0122] The urine collection receptacle **210** may also include a discharge port **220** at a lower end thereof (e.g., opposite the upper end). The discharge port **220** and associated mechanisms described below may allow for controlled emptying of the urinary fluid **214** from the urine collection receptacle **210**, enhancing the overall usability of the wearable urinary collection apparatus **200**.

[0123] In some implementations, the discharge port **220** may include an open-close mechanism **238**. The open-close mechanism **238** may comprise a small rubber ring band wrapped around a key

ring. A cord **240** may be attached to the key ring for balancing the pulling weight of the urinary fluid **214** accumulated within the urine collection receptacle **210**. For example, as the urinary fluid **214** collects in the urine collection receptacle **210**, the weight distribution of the wearable urinary collection apparatus **200** may change, potentially causing discomfort or altering the fit of the wearable urinary collection apparatus **200** or portions thereof. Accordingly, the cord **240** may enable the user to adjust the position of the urine collection receptacle **210**, ensuring comfort and proper alignment throughout use. Furthermore, the cord **240** may enhance the accessibility of the discharge port **220** by, for example, enabling users to locate and manipulate the open-close mechanism **238** more easily. This feature is particularly beneficial for users with mobility limitations or those who require assistance in managing the device. In some implementations, the urine collection receptacle **210** may include a cord aperture **242** through which the cord **240** may extend to further support the cord **240** with respect to the urine collection receptacle **210**.

[0124] As best shown in FIG. **16A**, the lower portion of the urine collection receptacle **210** may taper to the discharge port **220**. The tapering of the lower portion of the urine collection receptacle **210** towards the discharge port **220** may facilitate the controlled emptying of the urine collection receptacle **210**. This tapering may help direct the flow of urine towards the discharge port when the user is ready to empty the urine collection receptacle **210**. Where the urine collection receptacle **210** has a shorter length, the tapering of the lower portion of the urine collection receptacle **210** may not be necessary to direct the flow of the urinary fluid **214** to the discharge port **220**, and accordingly may not be included.

[0125] FIG. **17** is an upper perspective view illustration of another example of the wearable urinary collection apparatus **200** according to the present disclosure. The wearable urinary collection apparatus **200** may include variations and additional features as compared to the implementation shown in FIG. **15**. For example, in the illustrated implementation, the wearable urinary collection apparatus **200** includes a contractile ring **248** rather than the contractile stem tube **226**. The contractile ring **248** may be disposed within the tubular port **224** of the urine collection receptacle **210** (e.g., disposed centrally along a length of the tubular port **224**). The contractile ring **248** may provide an alternative means of creating a seal between the inner collection member **212** (e.g., the stream guide tube **234**) and the urine collection receptacle **210** at the interface **276**. This configuration may offer benefits such as easier assembly or improved sealing in certain situations. The contractile ring **248** may be made of a flexible, elastic material that may conform to the shape of the stream guide tube **234**, creating a tight seal to prevent leaks and backflow. Similar to the contractile stem tube **226**, the contractile ring **248** may have a relaxed inner diameter that is smaller than a relaxed outer diameter of the stream guide tube **234** such that the contractile ring **248** is configured to form a constriction along the stream guide tube **234** that functions similarly to the constriction formed from the contractile stem tube **226**.

[0126] The use of the contractile ring **248**, as opposed to the tubular structure of the contractile stem tube **226**, may offer several advantages. Firstly, the contractile ring **248** may streamline the assembly process of the wearable urinary collection apparatus **200** by allowing for easier insertion of the stream guide tube **234**, potentially reducing the time and complexity involved in assembling the wearable urinary collection apparatus **200**. This could be particularly beneficial in clinical settings where rapid assembly might be necessary. Additionally, the contractile ring **248** may provide improved sealing performance in certain situations, for example, by concentrating pressure at a specific point along the stream guide tube **234** to form more localized and potentially stronger seal at the interface **276**. This could be especially advantageous in ambulatory scenarios where the user's movements or changes in body position might put stress on the connection between the inner collection member **212** and the urine collection receptacle **210**.

[0127] FIG. **18** is an upper perspective view illustration of another example of the wearable urinary collection apparatus **200** according to the present disclosure. The wearable urinary collection apparatus **200** may include variations and additional features as compared to the implementation

shown in FIG. 15. For example, in the illustrated implementation, the wearable urinary collection apparatus **200** includes an outer adaptor ring **250** and an inner opening ring **252**. In some implementations, one or both of the outer adaptor ring **250** and the inner opening ring **252** may be formed from or comprise a semi-rigid material (e.g., an elastomeric rubber or the like). For example, one or both of the inner opening ring **252** and the outer adaptor ring **250** may be substantially compliant and flexible, such that, for example, the upper end of the redundant urine collection area **216** may be collapsable under clothes (e.g., a shirt) of the user. In some implementations, the inner opening ring **252** may include grooves and/or ridges applied circumferentially around the inversion **218** of the urine collection receptacle **210** configured for releasable connection with the outer adaptor ring **250** to form a secure attachment therebetween. The outer adaptor ring **250** may include fastening means to which the elongate mating members **222** may be connected. In such an implementation, the outer adaptor ring **250** may be configured to interface with (e.g., may include features for interfacing with) the elongate mating members **222**. For example, each of the elongate mating members **222** may include a loop **254** that is configured to extend around the outer adaptor ring **250**.

[0128] In some implementations, the inner opening ring **252** may be configured to maintain the shape of the inversion **218** and/or the redundant urine collection area **216** while providing a compliant (e.g., elastic) structure that may enhance user comfort. As described previously, in some implementations, the redundant urine collection area **216** of the urine collection receptacle **210** may be formed by inverting the upper portion of the urine collection receptacle **210** (e.g., an upper end of the urine collection receptacle **210**) over the inner opening ring **252** such that the inversion **218** is formed over the inner opening ring **252**. In such an implementation, the inversion **218** may be rounded as it extends over the inner opening ring **252**.

[0129] Where the inner opening ring **252** includes the grooves and/or ridges for connection with the outer adaptor ring **250**, the groove and/or ridges may be applied circumferentially around the inversion **218** of the urine collection receptacle **210**. These grooves and/or ridges may serve several functions, such as providing a mechanical interlock with the outer adaptor ring **250** to ensure a secure connection that may distribute the weight of the urine collection receptacle **210** evenly around the circumference of the inversion **218**. Additionally, the inner opening ring **252** may aid in maintaining the shape of the inversion **218**, which may be important for the proper functioning of the redundant urine collection area **216** (e.g., to prevent collapse of the redundant urine collection area **216**). In implementations where the inner opening ring **252** is disposed beneath the inversion **218** (e.g., where the redundant urine collection area **216** is formed by folding the upper end thereof over the inner opening ring **252**), the ridges and/or grooves may be so distinct as to be defined (e.g., connectable) through the material that forms the urine collection receptacle **210** (e.g., a TPU membrane).

[0130] In implementations where the urine collection receptacle **210** includes the inner opening ring **252**, the wearable urinary collection apparatus **200** may include a urine receptacle bag (not shown) rather than the urine collection receptacle **210**. In such an implementation, the urine receptacle bag may be configured to collect the urinary fluid **214** therein and may be formed as an independent member (e.g., rather than being formed by inverting the upper portion of the urine collection receptacle **210** inwards) that extends downward from the inner opening ring **252**. For example, the redundant urine collection area **216** may extend downward from the circumference of the inner opening ring **252**. The urine receptacle bag may also extend from the inner opening ring **252** (e.g., the circumference of the inner opening ring **252**) outward of the redundant urine collection area **216** and may extend fully around the redundant urine collection area **216** (e.g., may fully envelop the redundant urine collection area **216**). In some implementations, the urine receptacle bag and/or the redundant urine collection area **216** may be removably connectable to the inner opening ring **252** to form respective seals therebetween. The urine receptacle bag may be formed from the same material(s) as described previously with respect to the urine collection

receptacle **210**. Accordingly, this implementation of the wearable urinary collection apparatus **200** may function substantially similarly to other implementations of the wearable urinary collection apparatus **200** described herein.

[0131] FIGS. **19A-19B** are upper perspective view illustrations of other examples of the urine collection receptacle **210**. As shown, in some implementations, the urine collection receptacle **210** may include the inner opening ring **252** while excluding the outer adaptor ring **250**. In such implementations, the inner opening ring **252** may include mating features **244** that are configured to interface with the elongate mating members **222** to form a secure connection therebetween. In some implementations, the elongate mating members **222** may be configured to be removably coupleable to the inner opening ring **252** (e.g., via the mating features **244**) such that components of the wearable urinary collection apparatus **200** may be cleaned and/or replaced.

[0132] For example, in the implementation shown in FIG. **19A**, the inner opening ring **252** includes one or more apertures **244a** that are configured to receive the elongate mating members **222** (e.g., the loops **254** of the elongate mating members **222** shown in FIG. **18**) to secure the elongate mating members **222** to the urine collection receptacle **210**. For example, the inner opening ring **252** may be substantially cylindrical and may include two of the apertures **244a** on opposite ends of the inner opening ring **252**. The apertures **244a** may extend through the inner opening ring **252** from an inboard side to an outboard side thereof. As described previously, the redundant urine collection area **216** may be formed by folding the upper end of the urine collection receptacle **210** inward over the inner opening ring **252** such that the inner opening ring **252** is disposed below the inversion **218**. Accordingly, the urine collection receptacle **210** may include complementary apertures located adjacent to the one or more apertures **244a** such that the elongate mating members **222** are passable through the one or more apertures **244a**.

[0133] As another example, in the implementation shown in FIG. **19B**, the inner opening ring **252** includes one or more slots **244b** that, similarly to the apertures **244a**, are configured to receive the elongate mating members **222** (e.g., the loops **254** of the elongate mating members **222** shown in FIG. **18**) to secure the elongate mating members **222** to the urine collection receptacle **210**. For example, the inner opening ring **252** may comprise a thin tubular ring portion and tabular portions extending upward therefrom (e.g., away from the lower end of the urine collection receptacle **210**), and the one or more slots **244b** may extend through the tabular portions. For example, in the illustrated implementation, the inner opening ring **252** includes the thin tubular ring portion extending around a circumference of the redundant urine collection area **216** to support the opening of the redundant urine collection area **216**, and includes two of the tabular portions extending upwards from the thin tubular ring portion on opposite ends thereof. Furthermore, as shown, the inner opening ring **252** includes two of the slots **244b** that each extend through a respective one of the tabular portions (e.g., that each extend from an inboard side to an outboard side of a respective one of the tabular portions). By spacing the one or more slots **244b** away from the inversion **218**, the elongate mating members **222** may be inserted through the one or more slots **244b** without interference with the urine collection receptacle **210**. In some implementations, rather than the one or more slots **244b** that extend through respective tab portions, the inner opening ring **252** may include loops of material (not expressly shown) that extend therefrom and through which the elongate mating members **222** may extend. For example, the inner opening ring **252** may include two loops that extend upward from the inversion **218** at opposite ends of the inner opening ring **252**. As another example, the inner opening ring **252** may include two loops that extend outward from outboard sides of opposite ends of the inner opening ring **252**.

[0134] FIG. **20** is an upper perspective view illustration of another example the wearable urinary collection apparatus **200**. In some implementations, the redundant urine collection area **216** may not be formed from the urine collection receptacle **210** but rather may be a separate component. In such an implementation, each of the urine collection receptacle **210** and the redundant urine collection area **216** may be connectable to the inner opening ring **252**. For example, one or more of

the urine collection receptacle **210** or the redundant urine collection area **216** may be connected to the inner opening ring **252** via adhesives or the like. In some implementations, such as where the urine collection receptacle **210** and/or the redundant urine collection area **216** are formed from TPU membrane, one or more of the urine collection receptacle **210** or the redundant urine collection area **216** may be connected to the inner opening ring **252** by laminating the urine collection receptacle **210** and/or the redundant urine collection area **216** directly to the inner opening ring **252**. As shown, in the illustrated implementation as well as all other implementations described herein, the inner opening ring **252** may have a depth of about 2 cm.

[0135] As shown, to facilitate a connection with the inner opening ring **252**, the urine collection receptacle **210** may include first strips **270** that extend upward from an upper terminal edge of thereof. Similarly, to facilitate a connection with the inner opening ring **252**, the redundant urine collection area **216** may include second strips **272** that extend upward from an upper terminal edge thereof. Each of the first strips **270** and the second strips **272** may be configured to fold over the inner opening ring **252** to secure the urine collection receptacle **210** and the redundant urine collection area **216** to the inner opening ring **252**, respectively. For example, as depicted in FIG. **20** by bolded arrows, the first strips **270** of the urine collection receptacle **210** may be configured to extend upward along an outboard side of the inner opening ring **252** and fold inward over the inner opening ring **252** to secure the urine collection receptacle **210** to the inner opening ring **252**. Furthermore, also as depicted in FIG. **20** by bolded arrows, the second strips **272** of the redundant urine collection area **216** may be configured to extend upward along an inboard side of the inner opening ring **252** and fold outward over the inner opening ring **252** to secure the redundant urine collection area **216** to the inner opening ring **252**. In some implementations, the first strips **270** and/or the second strips **272** may be secured to the inner opening ring **252** via an adhesive or may be laminated directly to the inner opening ring **252**.

[0136] In some implementations of the wearable urinary collection apparatus shown in FIG. **20**, the inner opening ring **252** may be excluded, and the redundant urine collection area **216** may be connected directly to the urine collection receptacle **210**. For example, a circumference of an upper end of the redundant urine collection area **216** may be connected directly to a circumference of an upper end of the urine collection receptacle **210** that is complementary to the upper end of the redundant urine collection area **216**. For example, the circumference of the upper end of the redundant urine collection area **216** may be adhered and/or laminated directly to the circumference of the upper end of the urine collection receptacle **210** to form a seal therebetween. In such an implementation, the urine collection receptacle **210** may include the first strips **270**, in which the first strips **270** are configured to extend along an outboard side of the urine collection receptacle **210** and fold inward over the upper end of the redundant urine collection area **216** to secure the urine collection receptacle **210** to the redundant urine collection area **216**. In addition to, or in the alternative, the redundant urine collection area **216** may include the second strips **272**, in which the second strips **272** are configured to extend along an inboard side of the urine collection receptacle **210** and fold outward over the upper end of the urine collection receptacle **210** to secure the redundant urine collection area **216** to the urine collection receptacle **210**.

[0137] FIGS. **21A-21B** are upper perspective view illustrations of a harness **256** for the wearable urinary collection apparatus **200** according to an example. The harness **256** may be configured to support the urine collection receptacle **210** relative to a user. In some implementations, the harness **256** may be configured to support the urine collection receptacle **210** relative to a user such that the upper end (e.g., the inversion **218**, the inner opening ring **252**, and/or the outer adaptor ring **250**) of the urine collection receptacle **210** is near, but spaced from the user (e.g., a pelvic area of the user), and the lower end of the urine collection receptacle **210** extends away from the user (e.g., downward). As explained in further detail below, the harness **256** may be configured to support the urine collection receptacle **210** relative to the user without loosening or slipping (e.g., when the user changes postures). For example, the harness **256** may be configured to support the urine

collection receptacle **210** relative to the user without loosening or slipping (e.g., without substantial movement of the upper end of the urine collection receptacle **210** relative to the user) when the user moves from a seated position to a standing position and/or when the user moves from the standing position to the seated position. Furthermore, the harness **256** may be configured to support the urine collection receptacle **210** relative to the user such that the urine collection receptacle **210** clears (e.g., avoids interfering with) a seat (not shown) on which the user may sit. For example, the harness **256** may be configured to support the urine collection receptacle **210** relative to the user such that the urine collection receptacle **210** does not interfere with (e.g., contact) a toilet on which the user may sit. In such an implementation, for example, the harness **256** may support the urine collection receptacle **210** relative to the user such that the urine collection receptacle **210** is spaced inward (e.g., inside) or outwards (e.g., outside) of a bowl of the toilet.

[0138] In some implementations, the harness **256** may include securing straps **258** that extend around portions of the user to secure the harness **256** to the user. The securing straps **258** may extend from (e.g., through) either side of a padded member **264** and may be configured to extend around respective portions of the user, such as legs and/or the iliac crest of the buttocks, to support the urine collection receptacle **210** relative to the user. In some implementations, the securing straps **258** may further extend through apertures disposed on a bottom portion of the padded member **264** to form a loop that extends around the respective portions of the user. The apertures may be laterally spaced from each other to space the securing straps **258** outward of the penis **202** of the user. The padded member **264** may be a cylindrical (e.g., rectangular when flat) component positioned centrally between the securing straps **258** such as to lay substantially across a lap of the user when the harness **256** is worn by the user. The padded member **264** may be configured to allow the harness **256** to sit comfortably against the user's body, typically in the lower abdominal or pelvic area, when the urine collection receptacle **210** is connected to the harness **256**.

[0139] In some implementations, the harness **256** may include a waist strap **266** that passes through the padded member **264** and forms a loop configured to extend around a waist of the user. A fastening mechanism **268** may be positioned along the waist strap **266** to releasably connect opposing ends thereof. The waist strap **266** may be configured to further support the urine collection receptacle **210** relative to the user, for example, where the user is large and/or obese. Adjustment mechanisms **262** may be provided on the securing straps **258** to allow for size adjustment. The harness **256** may include complementary mating features **260** arranged vertically along strips extending downward from the padded member **264** for coupling the urine collection receptacle **210** to the harness **256** (e.g., for connection with the mating features **246** of the elongate mating members **222**). In some implementations, the strips may be connected to the padded member **264**. In other implementations, however, the strips may extend along (e.g., may be formed along) the securing straps **258** (e.g., portions of respective ones of the securing straps **258** proximate the groin and/or pelvic area of the user). Accordingly, in the illustrated implementation, the complementary mating features **260** of the harness **256** extend along portions of the securing straps **258** (e.g., portions of the securing straps **258** proximate the groin and/or pelvic area of the user). In some implementations, the complementary mating features **260** may comprise Velcro, small loops or hooks, pegs, snaps, or the like. The complementary mating features **260** may be removably couplable to the mating features of the urine collection receptacle **210** to removably connect the urine collection receptacle **210** to the harness **256**. In some implementations, the waist strap **266** may extend through the padded member **264** and work in conjunction with the securing straps **258** to provide support for the wearable urinary collection apparatus **200**. Furthermore, the harness **256** may work in conjunction with other components of the wearable urinary collection apparatus **200**, such as the elongate cord **240**, to provide a comprehensive solution for urinary collection and support.

[0140] FIG. **22** illustrates a method **300** for affixing the wearable urinary collection apparatus **200** to a user having a penis (e.g., the penis **202**). The method **300** may include several steps for

assembling and using the wearable urinary collection apparatus **200**.

[0141] The method may include inserting the penis **202** of the user into the inner collection member **212**. In some implementations, the penile sleeve **230** of the inner collection member **212** may be unrolled over the shaft of the penis **202**, similar to the application of a condom. This action helps to create a secure seal and prevents leakage at the base of the penis **202**. The voiding area **232** may be positioned adjacent to the penis glans **206**, aligning with the urethral opening **208** (e.g., urethra) with the tubular port **224** to efficiently capture the urinary fluid **214** as it is voided.

[0142] The length of penis **202** inserted into the inner collection member **212** may vary depending on the individual user's anatomy and comfort level. However, it is generally recommended to insert enough length to ensure that the stream guide tube **234** is properly positioned relative to the urethral opening **208** (e.g., urethra). This positioning may be important for directing urine flow into the urine collection receptacle **210** and minimizing the risk of leakage or backflow.

[0143] The method **300** may also include inserting **304** the one-way valve **236** of the inner collection member **212** through the tubular port **224** (e.g., the contractile stem tube **226**) of the urine collection receptacle **210**. In some implementations, inserting **304** the one-way valve **236** through the contractile stem tube **226** may form a constriction along the stream guide tube **234**. The one-way valve **236** may be disposed within the urine collection receptacle **210** after insertion through the contractile stem tube **226**.

[0144] In some implementations, inserting **304** the one-way valve **236** of the inner collection member **212** through the contractile stem tube **226** of the urine collection receptacle **210** may further comprise deforming the one-way valve **236** to reduce an axial size of the one-way valve **236** such that the one-way valve **236** is insertable through the contractile stem tube **226**.

[0145] This step in the assembly process highlights an important consideration when inserting the one-way valve **236** through the contractile stem tube **226**. The one-way valve **236**, being a flexible component, may need to be temporarily deformed or compressed to facilitate its passage through the contractile stem tube **226**. This deformation may involve reducing the cross-sectional size (e.g., an outer diameter) of the one-way valve **236**, which may be achieved by gently folding or compressing the one-way valve **236**.

[0146] The formation of a constriction along the stream guide tube **234** during the inserting **304** serves as an additional safeguard against backflow. The contractile nature of the tubular port **224** and/or the contractile stem tube **226** allows one or both of these components to grip the stream guide tube **234** tightly, creating a dynamic seal that may adapt to changes in pressure or movement. Once inserted, the one-way valve **236** is positioned within the urine collection receptacle **210**. This placement ensures that any urine that passes through the one-way valve **236** is contained within the main body of the urine collection receptacle **210**, further minimizing the risk of leakage or backflow. Furthermore, positioning of the one-way valve **236** inside the urine collection receptacle **210** may protect the one-way valve **236** from external forces that might compromise its function.

[0147] The method **300** may also include inverting **306** the upper end of the urine collection receptacle **210** that surrounds the contractile stem tube **226** inwards at the inversion **218** to form the redundant urine collection area **216**. In some implementations, the urine collection receptacle **210** may extend longitudinally from a lower end to terminate at the inversion **218**. The inverting action may involve folding the upper end of the urine collection receptacle **210** inward at a specific point, such as the inversion **218** or the inner opening ring **252**. This creates a double-walled section at the upper portion of the urine collection receptacle **210**, which serves multiple purposes as described previously.

[0148] The method **300** may also include inserting **308** the inner collection member **212** into the redundant urine collection area **216** of the urine collection receptacle **210**. Inserting the inner collection member **212** into the redundant urine collection area **216** of the urine collection receptacle **210** establishes the primary interface between the user's anatomy and the urine collection receptacle **210**. The inclusion of the contractile stem tube **226** within the redundant urine collection

area **216** is an important feature of the configuration. As the inner collection member **212** is inserted, its stream guide tube **234** passes through the contractile stem tube **226**. This creates a secure pathway for urine to flow from the user into the main body of the urine collection receptacle **210**.

[0149] The redundant urine collection area **216** also accommodates any minor leaks that might occur at the interface **276** between the stream guide tube **234** and the contractile stem tube **226**. This fail-safe feature further enhances the reliability of the wearable urinary collection apparatus **200**, ensuring that any potential leaks are contained within the system rather than reaching the user or their clothing.

[0150] The method **300** may also include coupling **310** mating features extending from the urine collection receptacle **210** to the complementary mating features **260** disposed along securing straps **258** of the harness **256**. The placement of the complementary mating features **260** along the securing straps **258** provides flexibility in positioning the urine collection receptacle **210** relative to the user (e.g., to increase or decrease the spacing between the inversion **218** and the user).

[0151] This method **300** may provide a systematic approach to assembling and affixing the wearable urinary collection apparatus **200**, ensuring proper alignment and connection of all components. The steps of the method **300** may be configured to maximize the effectiveness of the backflow prevention mechanisms and to ensure a secure and comfortable fit for the user.

[0152] FIG. **23** is an upper perspective view illustration of a wearable urinary collection system **400**. The wearable urinary collection system **400** may be similar to the wearable urinary collection apparatus **200** except as described herein. For example, the wearable urinary collection system **400** includes a fluid collection receptacle **410** that may be similar to the urine collection receptacle **210** and includes an inner member **412** that may be similar to the inner collection member **212**.

[0153] Additionally, like the urine collection receptacle **210**, the fluid collection receptacle **410** may be formed from or comprise a tubular membrane (e.g., a tubular sheet of polymer such as TPU membrane, etc.) as described previously. For example, similar to the wearable urinary collection apparatus **200**, the fluid collection receptacle **410**, the inner member **412** of the wearable urinary collection system **400**, and/or components thereof, may comprise or be formed from a thin membrane that is generally impermeable to and resistant to the urinary fluid **214**. For example, in some implementations, the fluid collection receptacle **410**, the inner member **412**, and/or components thereof may comprise or be formed from a thermoplastic polyurethane (TPU) membrane. Accordingly, the fluid collection receptacle **410** may be referred to herein as a tubular membrane.

[0154] Furthermore, similar to the inner collection member **212**, the inner member **412** may comprise an upper ring **428** that is similar to the upper band **228**, a shaft portion **430** that is similar to the penile sleeve **230**, and an end portion **432** that is similar to the voiding area **232**.

Accordingly, the inner member **412** may be referred to herein as a sleeve. The inner member **412** also includes a guide tube **434** that may be similar to the stream guide tube **234** and that is configured to interact with one or more portions of the fluid collection receptacle **410** to inhibit the urinary fluid **214** from backflowing (e.g., leaking) therefrom. The guide tube **434** may extend from an apex of the end portion **432**. In some implementations, the end portion **432** may be configured to provide a gap inside which the urinary fluid **214** immediately discharged from the penis **202** may collect before traveling through the guide tube **434**. However, in other implementations, such as the implementation shown in FIG. **23**, the end portion **432** may be configured to leave no such gap, in which the urethral opening **208** (e.g., urethra) of the penis **202** is configured to be positioned immediately adjacent to the guide tube **434** (e.g., at an opening of the guide tube **434**) such that the urinary fluid **214** immediately discharged from the penis **202** is directed immediately into the guide tube **434**.

[0155] Finally, like the urine collection receptacle **210**, the fluid collection receptacle **410** of the wearable urinary collection system **400** may include a redundant fluid collection portion **416** that is

similar to the redundant urine collection area **216** of the urine collection receptacle **210**. As explained herein, the redundant fluid collection portion **416** may be configured to receive the inner member **412**, or a portion thereof, such that the redundant fluid collection portion **416** extends around at least a portion of the inner member **412**. Other components of the wearable urinary collection system **400** may also be similar to those of the wearable urinary collection apparatus **200** although not explicitly recited herein.

[0156] The fluid collection receptacle **410** is configured to cooperate with the inner member **412** to direct the urinary fluid **214** from the penis **202** to be collected within the fluid collection receptacle **410** and to prevent backflow (e.g., leaking) of the urinary fluid **214** therefrom. Furthermore, the fluid collection receptacle **410** may include a series of seals **424** that are configured to inhibit (e.g., cooperatively inhibit) the urinary fluid **214** collected within the fluid collection receptacle **410** from traveling toward the redundant fluid collection portion **416**. For example, the guide tube **434** of the inner member **412** may be configured to extend through some of the seals **424** and terminate between adjacent ones of the seals **424**. Accordingly, a fluid flow path may be formed from the inner member **412**, through the guide tube **434**, through apertures of remaining of the seals **424**, and into a volume defined by the fluid collection receptacle **410**. Because the fluid flow path extends through the remaining of the seals **424**, the urinary fluid **214** collected in the volume may be inhibited from traveling (e.g., from traveling upward) toward the redundant fluid collection portion **416**.

[0157] For example, in the illustrated implementation, each of the seals **424** extend along a width of the fluid collection receptacle **410** (e.g., between lateral sides of the fluid collection receptacle **410**). The seals **424** are spaced along a length of the fluid collection receptacle **410** that is generally perpendicular (e.g., orthogonal) to the width, such as to define redundant collection chambers **442** between adjacent pairs of the seals **424**. Each of the seals **424** may also include an aperture **426** that extends therethrough and that may fluidly connect adjacent ones of the redundant collection chambers **442**. Each of the redundant collection chambers **442** may be defined by two of the seals **424** (e.g., adjacent ones of the seals **424**) and may be substantially sealed from others of the redundant collection chambers **442**, the volume defined by the fluid collection receptacle **410** for containing the urinary fluid **214**, the redundant fluid collection portion **416**, and/or an environment around the fluid collection receptacle **410**.

[0158] Each of the redundant collection chambers **442** is configured to collect the urinary fluid **214** that travels from the fluid collection region through the apertures **426** of one or more of the seals **424**. As shown, the redundant collection chambers **442** may be arranged in series with respective apertures **426** coupling adjacent ones of the redundant collection chambers **442**. In some implementations, the redundant collection chambers **442** may be sealed from respective adjacent ones of the redundant collection chambers **442** except through the respective apertures **426**. Accordingly, the series of the seals **424**, the apertures **426**, and the redundant collection chambers **442** may be operable (e.g., cooperable) to prevent backflow out of the fluid collection receptacle **410**. For example, the redundant collection chambers **442** may be configured to capture any of the urinary fluid **214** that travels (e.g., leaks) past the seals **424**. Accordingly, the urinary fluid **214** may travel through the guide tube **434** and/or the apertures **426**, but may inadvertently leak past the outer surface of the guide tube **434** and the redundant collection chambers **442** may be positioned to collect this escaped fluid, adding an extra layer of protection against leakage.

[0159] The size of the redundant collection chambers **442** may vary depending on the specific design and anticipated leakage volume. In some implementations, the redundant collection chambers **442** may be relatively small, just large enough to contain a minimal amount of leaked fluid. In other implementations, the redundant collection chambers **442** may be larger to accommodate more substantial leakage, such as in scenarios where the seals **424** are subjected to high pressure or frequent movement. The arrangement of the redundant collection chambers **442** in series, with respective apertures **426** coupling adjacent ones of the redundant collection chambers

442, may create a tortuous path for the urinary fluid **214**, further hindering backflow. In some implementations, the redundant collection chambers **442** (or at least a lower redundant collection chamber **442b**) may be fluidly coupled to the volume defined by the fluid collection receptacle **410**. This may be achieved through one or more apertures **426** (e.g., a lower aperture **426b** of a lower one of the seals **424**) which may allow any fluid collected in the redundant collection chambers **442** to drain back into the volume defined by the fluid collection receptacle **410**.

[0160] In some implementations, each of the redundant collection chambers **442** may be further defined by an upper portion of the fluid collection receptacle **410** (e.g., a wall of the upper portion of the fluid collection receptacle **410**). For example, similar to the urine collection receptacle **210**, the redundant fluid collection portion **416** may be formed by inverting (e.g., folding) the upper portion of the fluid collection receptacle **410** inwards at an inversion (e.g., a rim **418**). In addition to the redundant fluid collection portion **416**, the series of the seals **424** may be positioned on the upper portion of the fluid collection receptacle **410** and may be disposed within the fluid collection receptacle **410** (e.g., the volume defined thereby) when the upper portion is inverted (e.g., folded) inwards. Stated differently, the upper portion of the fluid collection receptacle **410** may include the series of seals **424** and the redundant fluid collection portion **416** may be formed from the upper portion of the tubular membrane folding inwards at the rim **418** such that the tubular membrane extends longitudinally from the rim **418** to a lower portion that is opposite the upper portion.

[0161] Accordingly, when the upper portion of the fluid collection receptacle **410** is inverted (e.g., folded) inwards such as to form the redundant fluid collection portion **416** and to dispose the series of the seals **424** within the volume of the fluid collection receptacle **410** (as shown in FIG. 23), the fluid collection receptacle **410** may define an upper region **448** that includes the rim **418** and the redundant fluid collection portion **416**, a backflow prevention region **450** that includes the series of the seals **424** disposed within the volume of the fluid collection receptacle **410**, and a fluid collection region **452** that includes the remainder of the volume of the fluid collection receptacle **410** and which is primarily configured to collect the urinary fluid **214**. Accordingly, in some implementations, the backflow prevention region **450** may be disposed between the upper region **448** and the fluid collection region **452**.

[0162] The fluid collection receptacle **410** may include any number of the seals **424**. For example, in the illustrated implementation, the fluid collection receptacle **410** includes eight of the seals **424** (e.g., accordingly defining seven of the redundant collection chambers **442**). Accordingly, the series of the seals **424** may include an upper seal from the seals **424** and a lower seal from the seals **424**, in which the upper seal includes the upper aperture **426a** and the lower seal includes the lower aperture **426b**. Furthermore, the upper seal and one of the seals **424** adjacent to the upper seal may define an upper redundant collection chamber **442a** and the lower seal and one of the seals **424** adjacent to the lower seal may define a lower redundant collection chamber **442b**. In other implementations, however, the fluid collection receptacle **410** may include any other number of the seals **424**. For example, the fluid collection receptacle **410** may include two of the seals **424**, three of the seals **424**, five of the seals **424**, ten of the seals **424**, sixteen of the seals **424**, etc.

[0163] In some implementations, the seals **424** may be configured to provide support and stability to the wearable urinary collection system **400**. For example, the seals **424** may be configured to prevent the guide tube **434** from slipping relative to the apertures **426** and to maintain a shape of the fluid collection receptacle **410**. Furthermore, the number (e.g., the quantity) of the seals **424** may be dependent upon the dimensions of the fluid collection receptacle **410** and the number of the seals **424** required to adequately support the wearable urinary collection system **400** (e.g., without kinking, twisting, etc.). To further support the fluid collection receptacle **410**, one or more of the seals **424** may extend beyond the redundant fluid collection portion **416** to the outer wall of the fluid collection receptacle **410**. In other words, at least one of the seals **424** may include bridges **456** that extend between the redundant fluid collection portion **416** and the fluid collection receptacle **410** along the backflow prevention region **450**. For example, in the illustrated

implementation, the upper-most of the seals **424**, the third one of the seals **424** from the upper-most, and the fifth one of the seals **424** from the upper-most include the bridges **456** that extend between the wall forming the redundant fluid collection portion **416** and the outer wall of the fluid collection receptacle **410**. By including the bridges **456** with some of the seals **424**, the redundant fluid collection portion **416** may be supported relative to the outer wall of the fluid collection receptacle **410** such that the redundant fluid collection portion **416** and the series of the seals **424** are generally inhibited from moving relative to the outer wall of the urine collection receptacle **410**. [0164] When the inner member **412** is inserted into the redundant fluid collection portion **416** of the fluid collection receptacle **410**, the guide tube **434** of the inner member **412** is configured to extend through the apertures **426** of at least some of the seals **424** to form a unidirectional fluid flow path from the penis **202** into the volume defined by the fluid collection receptacle **410**. For example, a unidirectional fluid flow path for the urinary fluid **214** may extend from the penis **202** into the inner member **412**, then into the guide tube **434**, and then through the apertures **426** of any remaining ones of the seals **424** and into the fluid collection region. Stated differently, when the redundant fluid collection portion **416** extends around the inner member **412**, or a portion thereof, the guide tube **434** may extend at least through the upper redundant collection chamber **442a** by extending through the upper aperture **426a** of the upper seal to terminate within one of the redundant collection chambers **442** of the backflow prevention region **450**. In some implementations, the guide tube **434** may be configured to extend through the apertures **426** of more than half of the seals **424**. In other implementations, the guide tube **434** may be configured to extend through the apertures **426** of less than half of the seals **424**. In the illustrated implementation, a terminal end **436** of the guide tube **434** is shown to be configured to be disposed within one of the redundant collection chambers **442** that is between a fifth of the seals **424** and a sixth of the seals **424**. However, in other implementations, the terminal end **436** may be configured to be disposed within a different one of the redundant collection chambers **442**.

[0165] In some implementations, the terminal end **436** of the guide tube **434** may be positioned within the upper redundant collection chamber **442a**, the lower redundant collection chamber **442b**, or any other of the redundant collection chambers **442**. Where the terminal end **436** of the guide tube **434** is configured to be positioned within the lower redundant collection chamber **442b**, the urinary fluid **214** may flow from the guide tube **434**, into the lower redundant collection chamber **442b**, through the lower aperture **426b** of the lower seal, before flowing into the fluid collection region **452**. Accordingly, the lower redundant collection chamber **442b** may serve as an intermediary collection point for the urinary fluid **214** before the urinary fluid **214** enters the fluid collection region **452** for collection.

[0166] By including the seals **424** through which the guide tube **434** does not extend, those remaining ones of the seals **424** (e.g., and the apertures **426** thereof) may be configured to inhibit (e.g., prevent) the urinary fluid **214** collected in the volume defined by the fluid collection receptacle **410** from traveling from the fluid collection region **452**, through the backflow prevention region **450**, and to the upper region **448**. Stated differently, the series of the seals **424** may be configured to inhibit the urinary fluid **214** collected in the fluid collection receptacle **410** from traveling upwards to the redundant fluid collection portion **416**.

[0167] The series of seals **424** may be formed by sealingly connecting (e.g., adhering, bonding, etc.) opposite sides of the fluid collection receptacle **410** (e.g., opposite lateral sides of the upper portion of the fluid collection receptacle **410**). For example, where the fluid collection receptacle **410** is formed from a thin polymer film (e.g., TPU), the seals **424** may be formed by laminating opposite lateral sides of the fluid collection receptacle **410**. In some implementations, each of the seals **424** may be formed to include overlapping layers (e.g., flaps) at the respective one of the apertures **426**. Each of the flaps may be configured substantially similarly to the one-way valve **236** of the inner collection member **212**. Accordingly, each of the apertures **426** may define an opposing wall and a flap that is biased (e.g., naturally biased) toward the opposing wall to inhibit the urinary

fluid **214** from traveling upwards (e.g., toward the penis **202**) therethrough. However, when the urinary fluid **214** travels (e.g., flows) downwards (e.g., away from the penis **202**) through the respective one of the apertures **426**, the urinary fluid **214** may urge the flap away from the opposing wall to enable the urinary fluid **414** to travel therethrough. Similarly, the guide tube **434** may also be configured to urge the flap away from the opposing wall when the guide tube **434** extends through the respective one of the apertures **426** (e.g., when positioning the guide tube **434** within the backflow prevention region **450**).

[0168] When the guide tube **434** is positioned within the apertures **426** (e.g., of at least some of the seals **424**), the guide tube **434** may be configured to interface with respective ones of the apertures **426** to inhibit the urinary fluid **214** from traveling around the guide tube **434** through the apertures **426**. For example, an outer surface of the guide tube **434** may be configured to sealingly interface with respective ones of the apertures **426**. Furthermore, in some implementations, each of the apertures **426** may be configured to form a respective constriction along the guide tube **434** (e.g., similar to the contractile ring **248**) that inhibits the urinary fluid **214** from traveling from the volume defined by the fluid collection receptacle **410** through the guide tube **434** to the end portion **432** of the inner member **412**.

[0169] In some implementations, the guide tube **434** may be sealed within the apertures **426** of some of the seals **424** to establish a fluid-tight connection to, for example, ensure that the urinary fluid **214** is unable to travel (e.g., leak) through the interfaces between the guide tube **434** and the respective apertures **426**, maintaining a controlled fluid flow path. This sealing can be achieved through various methods, including heat sealing, bonding with a biocompatible adhesive (e.g., hydrogel), or mechanical constriction (e.g., one or more of the contractile stem tube **226** or the contractile ring **248**). For example, a diameter of the guide tube **434** may be substantially equal to a diameter of the apertures **426** of some of the seals **424**, ensuring a close fit that allows the guide tube **434** to pass through the apertures **426** without a clearance therebetween, minimizing the potential for leakage. Alternatively, the apertures **426** could be slightly smaller than the outer diameter of the guide tube **434**, creating an interference fit that forms a secure seal. In such an implementation, the seals **424** may be formed from an elastomeric polymer to enable the guide tube **434** to slideably and sealingly engage with the aperture **426**. In still other implementations, a bio-compatible sealant (e.g., hydrogel) may be used to establish a liquid seal between the aperture **426** and the guide tube **434**.

[0170] Stated differently, respective interfaces between the guide tube **434** and the aperture **426** of the seals **424** may be configured to inhibit the urinary fluid **214** collected in the fluid collection region **452** from traveling to the redundant fluid collection portion **416**. For example, the series of seals **424** and the guide tube **434** may be configured such that each of the redundant collection chambers **442** collect the urinary fluid **214** that travels from the fluid collection region through respective interfaces between the guide tube **434** and the aperture **426**.

[0171] The fluid collection receptacle **410** may also include a discharge port **420** that is similar to the discharge port **220**, including an open-close mechanism **438** and a cord **440** configured to selectively open and close the discharge port **420** to allow for removal (e.g., discharge) of the urinary fluid **214** from within the fluid collection receptacle **410** (e.g., to empty the fluid collection receptacle **410**).

[0172] Additionally, the fluid collection receptacle **410** may include one or more apertures **444** that are configured to interface with (e.g., receive) connecting straps **422** and that include one or more mating features **446** that are configured to interface with (e.g., engage, receive, etc.) complementary mating features of a harness (e.g., the complementary mating features **260** of the harness **256**). In some implementations, the mating features **446** are spaced around the rim **418** of the fluid collection receptacle **410** (e.g., spaced equidistantly around the rim **418**). The one or more apertures **444** may be similar to the mating features **244** and the connecting straps **422** may be similar to the elongate mating members **222**. The mating features **446** and the complementary

mating features **260** may be positioned to evenly distribute the weight of the fluid collection receptacle **410** and to prevent strain on any particular area of the body. As explained above with reference to FIGS. **21A-21B**, the harness **256** may be designed to be unobtrusive, allowing the user to wear it comfortably under clothing, and may include padding or cushioning in key areas to further enhance comfort and prevent irritation of the skin.

[0173] By spacing the one or more apertures **444** around the rim **418**, the connecting straps **422** may be attached to the fluid collection receptacle **410** in a manner that distributes the weight of the fluid collection receptacle **410** evenly. In some implementations, the one or more apertures **444** may include multiple of the apertures **444** that are spaced around (e.g., equidistantly spaced around) the rim **418** to, for example, distribute a load of the fluid collection receptacle **410** substantially evenly around the rim **418**. The apertures **444** may also be configured to enable the fluid collection receptacle **410** to be removed from the harness **256**. The quantity of the apertures **444** and the spacing between the apertures **444** may depend upon the size (e.g., diameter) of the fluid collection receptacle **410** (e.g., the rim **418**).

[0174] In some implementations, in addition to or in the alternative of a harness (e.g., the harness **256**), the wearable urinary collection system **400** may include a pouch that is configured to receive at least the fluid collection region **452** of the fluid collection receptacle **410**. The pouch may be configured to be secured closely to a leg of the user (e.g., may be configured to conform to the leg of the user) by, for example, straps that extend around the leg such that the fluid collection receptacle **410** is inhibited from moving away from the leg of the user, which may otherwise cause kinks in the guide tube **434** or other undesirable constrictions due to, for example, the weight of the urinary fluid **214** therein. In some implementations, the pouch may be formed integrally with the harness **256**, or may be formed with another piece of clothing. For example, the pouch may be formed with underwear, pants, belts, or some other article of clothing that is configured to be worn near the pelvis of a person.

[0175] FIGS. **24A** and **24B** illustrate the inner member **412** in both folded (FIG. **24A**) and unfolded (FIG. **24B**) configurations. FIG. **24A** is an upper perspective view illustration showing the inner member **412** in a rolled (e.g., stowed) configuration. As shown, the upper ring **428** is disposed at a top end of the inner member **412**, the shaft portion **430** is rolled around the upper ring **428**, and the guide tube **434** extends from a terminal end of the end portion **432** to terminate at the terminal end **436**. FIG. **24B** is an upper perspective view illustration of the inner member **412** in an unfolded (e.g., deployed) configuration. As shown, the shaft portion **430** is unrolled and extends downward from the upper ring **428**. To extend the inner member **412** around the penis **202**, the penis glans **206** may be positioned within the end portion **432** while the inner member **412** is in the rolled configuration (FIG. **24A**) such that the urethral opening **208** (e.g., urethra) of the penis **202** is positioned adjacent to the guide tube **434**. Thereafter, the shaft portion **430** may be unrolled over the shaft **204** of the penis **202** such that the upper ring **428** is disposed along the shaft **204** of the penis **202**. Hydrogel or the like may be used to secure the shaft portion **430** and/or the upper ring **428** to the shaft **204** of the penis **202**. In some implementations, additional securing features may be employed that connect the upper ring **428** to the fluid collection receptacle **410** (e.g., the rim **418** of the fluid collection receptacle **410**). For example, wires, strings, or the like may be secured between the rim **418** and the upper ring **428** to prevent the inner member **412** from sliding off of the penis **202**.

[0176] FIG. **25** illustrates a method **500** for affixing the wearable urinary collection system **400** to a user having a penis **202**. The method **500** may include inserting **502** the penis glans **206** of the penis **202** into the inner member **412**. For example, the method **500** may include unrolling the inner member **412** over the penis **202** such that the end portion **432** of the inner member **412** is adjacent to or over the penis glans **206** and the shaft portion **430** of the inner member **412** extends around the shaft **204** of the penis **202**.

[0177] As described previously, the fit between the inner member **412** and the penis **202** may be

selected to provide a secure, yet comfortable fit without constricting the penis **202** or inhibiting the flow of the urinary fluid **214**. The upper ring **428** of the inner member **412** may also aid in inhibiting the inner member **412** from slipping down the penis **202**. In some implementations, the inner member **412** may be lubricated prior to the inserting **502** of the penis glans **206** into the inner member **412** to reduce friction.

[0178] The method **500** may also include folding **504** an upper portion of the fluid collection receptacle **410** that includes the series of seals **424** inwards at the rim **418** to form the redundant fluid collection portion **416**. The fluid collection receptacle **410** may extend longitudinally from the rim **418** to a lower portion of the fluid collection receptacle **410**. As described previously, the series of redundant collection chambers **442** and the guide tube **434** may provide a tortuous path to inhibit backflow out of the fluid collection receptacle **410**. In some implementations, the upper portion of the fluid collection receptacle **410** may be folded inwards at the rim **418** prior to the assembly of the wearable urinary collection system **400** to save the user the need to fold the upper portion of the fluid collection receptacle **410** inwards at the rim **418**.

[0179] The method **500** may also include inserting the inner member **412** into the redundant fluid collection portion **416** of the fluid collection receptacle **410**. The method **500** may also include inserting **508** the guide tube **434** of the inner member **412** through some of the seals **424** via the apertures **426** to form interfaces between the guide tube **434** and the apertures **426** that inhibit the urinary fluid **214** from traveling from the fluid collection region **452** to the redundant fluid collection portion **416**.

[0180] The method **500** may also include coupling **510** mating features **446** extending from the fluid collection receptacle **410** to complementary mating features **260** disposed along the securing straps **258** of the harness **256**. The securing straps **258** are configured to extend around portions of the user to support the fluid collection receptacle **410** relative to the user. For example, the securing straps **258** may be configured to extend around the user's waist, legs, or a combination thereof, ensuring a secure and comfortable fit. The harness **256** provides a means for maintaining the fluid collection receptacle **410** in a stable position, preventing movement or displacement during activity. The securing straps **258** may be adjustable to accommodate different body sizes and shapes, and to optimize the position of the fluid collection receptacle **410** for comfort and effectiveness. The coupling **510** may involve a hook-and-loop fastener, a snap-fit mechanism, or any other means of attachment.

[0181] FIG. **26** is an upper perspective view illustration of another example of a wearable urinary collection system **600** according to the present disclosure including an inner sleeve **612** and a guide tube **632** disposed within a intermediate collection portion **616** of a Fluid collection receptacle **610**. The wearable urinary collection system **600** is similar to wearable urinary collection system **400** except as described herein.

[0182] The wearable urinary collection system **600** includes a fluid collection receptacle **610** that is similar in structure and function to the fluid collection receptacle **410** and is configured to contain urinary fluid **214** within a volume defined by the fluid collection receptacle **610**. The fluid collection receptacle **610** may comprise a thin, flexible polymer membrane **611** (e.g., thermoplastic polyurethane (TPU)) that is resistant and impermeable to urinary fluid **214**. The TPU film may have a thickness between 0.05 mm and 0.15 mm to provide sufficient flexibility while maintaining structural integrity under variable loading conditions. This thickness is selected to limit kinking and to allow the receptacle to conform to the user's anatomy. In some implementations, to reduce the vapor pressure in the volume defined by the fluid collection receptacle **610** that may result from the urinary fluid **214**, a salt and/or a salt solution (e.g., having characteristics similar to soy sauce) may be positioned within the volume such that the salt and/or the salt solution interacts with the urinary fluid **214**.

[0183] The wearable urinary collection system **600** also includes an inner sleeve **612** and may optionally include a separate guide tube **632**, rather than the integrated inner member **412** of the

wearable urinary collection system **400**. The inner sleeve **612** is configured to extend along the shaft **204** of the penis **202** while leaving the penis glans **206** exposed. The material of the inner sleeve **612** may be selected from elastic thermoplastic elastomers, such as natural rubber latex, polyisoprene, or polyurethane, to provide a snug but comfortable fit. These materials are commonly used in personal health and hygiene products due to their combination of softness, stretchability, and fluid impermeability. The material selection helps accommodate movement while minimizing the likelihood of crimping, bunching, or fluid leakage along the surrounding pathways.

[0184] The fluid collection receptacle **610** includes an intermediate collection portion **616** that is similar to the redundant fluid collection portion **416** of system **400**. The intermediate collection portion **616** may be formed by inverting an upper portion of the fluid collection receptacle **610** inward at a rim **618**, such that the upper portion is disposed within the internal volume defined by the receptacle **610**. The inversion at the rim **618** may define a double-walled configuration in which a folded-over upper segment **617** of the fluid collection receptacle **610** is received within the main body of the receptacle, thereby forming overlapping inner and outer layers along the intermediate collection portion **616**. This double-walled structure may enhance the structural integrity of the portion **616** and naturally resist deformation under external pressure, helping to preserve the internal shape and volume necessary for uninterrupted fluid collection and transfer.

[0185] A series of seals **622** may be provided along a length of the intermediate collection portion **616**, extending across its width, similar to seals **424** of the fluid collection receptacle **410**. These seals **622** may define redundant collection chambers **642** and may each include an aperture **626** that fluidly connects adjacent ones of the redundant collection chambers **642**. The apertures **626** may include a flap **627** that functions similarly to the one-way valve **236** of FIG. **18**, and may be axially aligned. The lowermost aperture **626** fluidly connects a lowermost chamber **642** to the main volume of the receptacle **610**.

[0186] The intermediate collection portion **616** also includes a port **636** that fluidly connects to the series of seals **622**. An upper seal **624** may be provided to define a lower wall of the intermediate collection portion **616** and may include the port **636**. The upper seal **624** may be frustoconical in shape to promote fluid guidance toward the port **636**. The tapering geometry of the frustoconical wall may also help to direct flow and reduce pooling, thereby minimizing the risk of backpressure that could result in retrograde leakage.

[0187] In some examples, one or more seals **622** may include respective bridges **644** that extend to the outer wall of the receptacle **610** to provide structural reinforcement between the inner and outer regions of the receptacle. These bridges **644** may include thickened or multilayered regions of bonded film, or embedded flexible supports, that limit axial collapse of the collection region when subjected to bending loads.

[0188] The guide tube **632** may be configured to extend through the port **636** and at least some of the apertures **626** of the seals **622**, defining a flow path from the intermediate collection portion **616** through the series of seals **622** and into the fluid collection region **652** of the receptacle **610**. The guide tube **632** may include upper and lower terminal ends **634**, with the upper end **634a** disposed near the port **636** and the lower end **634b** within one of the redundant collection chambers **642**. The guide tube **632** may have any length such as to extend through any number of the seals **622** (e.g., through any number of the apertures **626** and/or the redundant collection chambers **642**). As shown, the guide tube **632** has a length to extend through the port **636** and four of the seals **622**; however, the guide tube **632** may have any length such as to extend through more or fewer of the seals **622**.

[0189] The guide tube **632** may be formed from an elastomeric rubber material with a sufficient wall thickness (e.g., 0.75-1.25 mm) to maintain patency while resisting kinking or collapse, even under dynamic body movement or external compression. The guide tube **632** may be reinforced with embedded helical or ring structures **633** formed from semi-rigid thermoplastics, or may include ribbed outer surfaces to further enhance resistance to occlusion. Similar to the guide tube

434 of the inner member **412**, the guide tube **632** may be configured to interface with respective ones of the apertures **626** to form a seal therebetween. For example, the apertures **626** may have a relaxed inner diameter that is smaller than a relaxed outer diameter of the guide tube **632** such that when the guide tube **632** is inserted through the apertures **626**, respective interference fits therebetween may inhibit the urinary fluid **214** from traveling between adjacent ones of the redundant collection chambers **642** through the interface between the guide tube **632** and the apertures **626**.

[0190] In some implementations, one or more of the apertures **626** through which the guide tube **632** is configured to extend may include or be formed from an elastomeric material (e.g., similar to the contractile ring **248**) that is configured to form respective constrictions along the guide tube **632**. The respective constrictions may be configured to inhibit the urinary fluid **214** from traveling upwards through the guide tube **632** and may be configured to allow the urinary fluid **214** to travel downwards through the guide tube **632** (e.g., may promote unidirectional flow of the urinary fluid **214** through the guide tube **632**). In some implementations, to prevent the guide tube **632** from moving relative to the apertures **626** (e.g., axially twisting, etc.), the guide tube **632** may include adhesive tape disposed around an outer surface thereof that secures the guide tube **632** to the apertures **626**. Furthermore, a larger interference fit (e.g., by employing a guide tube having a larger diameter) may further inhibit movement of the guide tube **632** relative to the apertures **626**.

[0191] In one example implementation of system **600**, the guide tube **632** may be omitted and a short open channel or a small fluid discharge opening may be provided instead thereof. The opposing inner surfaces of the membrane forming the intermediate collection portion **616** may be bonded to one another at a transverse location to define a seal line, directing fluid flow downward through the open channel. Below this sealed region, the remaining wall of the tubular membrane may include a pair of flaps formed by cutting the corners on the left and right side, such that the lower portion of the wall forms two opposing flaps. These flaps may passively fold inward or outward in response to fluid flow. When urinary fluid flows downward by gravity, the flaps open to permit passage. If backpressure develops or fluid attempts to reverse direction, the flaps collapse inward and overlap, thereby blocking reverse flow. This passive backflow prevention feature may serve as a simple and cost-effective alternative to conventional valves.

[0192] In another variant of system **600**, a flap structure without the transverse sealing lines may be used. In this variant, the upper half of the flap region may include front and rear layers of the membrane that are fused together along their horizontal edges while remaining unfused along a vertical mid-section. This vertical region may define a narrow channel that may optionally accommodate a short guide tube or provide a direct stream path. The lower half of the flap structure includes a region where the front and rear walls are cut loose at the left and right corners, thereby forming opposing flaps similar to the Mushroom version. These flaps may respond dynamically to fluid flow conditions as previously described.

[0193] The inner sleeve **612** includes a sleeve portion **630** and one or more elastic bands **628**, such as those shown located at upper and lower terminal ends of the inner sleeve **612**. The inner sleeve **612** is configured to be worn such that the sleeve portion **630** extends along the shaft **204** of the penis **202** and the penis glans **206** remains exposed such that the urinary fluid **214** may be discharged freely from the urethral opening **208** into the intermediate collection portion **616** (e.g., without being blocked or diverted). The bands **628** may function similarly to the upper ring **428** of the inner member **412** and may be configured to inhibit the inner sleeve **612** from slipping along the shaft **204** of the penis **202**. For example, the inner sleeve **612** may be configured to apply a gentle radial pressure, such as about **200** mmHg per square inch, to maintain its position along the shaft **204** of the penis **202**. In some implementations, the a biocompatible adhesive (e.g., hydrogel, etc.) may be applied to the inner sleeve **612** (e.g., around the bands **628**) to further sure the inner sleeve **612** along the shaft **204**. The application of a biocompatible adhesive, such as hydrogel, may also seal the inner sleeve **612** along the shaft **204** of the penis **202** to further inhibit unintentional

travel (e.g., leaking) of the urinary fluid **214** during use.

[0194] The inner sleeve **612** is configured to be worn such that the penis glans **206** is exposed and positioned to discharge urine directly into the intermediate collection portion **616**. In use, the urinary fluid **214** may be discharged from the urethral opening **208** into the intermediate collection portion **616** and then flow downward through the port **636** into the fluid collection region **652**. The intermediate collection portion **616** may be approximately **10** cm wide and **10** cm long in a flat-lying state. Materials used for the sleeve **612** may include layered composites that combine soft elastomers for skin contact with stretchable but non-kinking textile fibers (e.g., warp-knit polyester) to maintain structural orientation during movement.

[0195] Accordingly, as shown, when the fluid collection receptacle **610** is worn on the user, the penis **202** may hang into the intermediate collection portion **616** such that the penis glans **206** is disposed centrally therein. Accordingly, the urinary fluid **214** that is discharged from the urethral opening **208** may fall freely into the intermediate collection portion **616** to drain through the port **636** and into the fluid collection region **652**. The inner sleeve **612** may be configured to apply a pressure of about 200 mmHg per square inch to the shaft **204** of the penis **202**.

[0196] In some configurations, a diaphragm **646** may extend inward from the rim **618** toward the sleeve **612**. The diaphragm **646** may include a central aperture **649** through which the inner sleeve **612** extends and may form a seal around the sleeve **612** to inhibit escape of urine from the intermediate collection portion **616**. The diaphragm **646** may have an outer diameter greater than that of the rim **618** to allow overlapping engagement. The diaphragm material may include a flexible, medical-grade silicone membrane or a TPU laminate selected to maintain elastic memory while conforming tightly to the contours of the sleeve **612**.

[0197] The diaphragm **646** may be sealingly connected to the rim **618** and may include a central aperture through which the inner sleeve **612** may extend. The diaphragm **646** may interface with the inner sleeve **612** (e.g., around a central region of the sleeve portion **630**) and may form a seal therebetween to inhibit the urinary fluid **214** that is collected in the intermediate collection portion **616** from escaping therefrom (e.g., through the central aperture of the diaphragm **646**). The diaphragm **646** may have an outer diameter that is larger than a diameter of the rim **618** such that material (e.g., a skirt) of the diaphragm **646** may overlap and/or interface with the rim **618** to form a seal therebetween. When the inner sleeve **612** is inserted through the central aperture of the diaphragm **646**, the inner sleeve **612** may extend proud of the diaphragm **646** to a length of about 3 cm. In other implementations, the inner sleeve **612** may terminate at the central aperture of the diaphragm **646** such that the inner sleeve **612** is disposed below the diaphragm **646**.

[0198] Similar to the fluid collection receptacle **410**, the fluid collection receptacle **610** may include a discharge port **620** that is similar to the discharge port **420**, and that may include an open-close mechanism **638** and a cord **640**. The open-close mechanism **638** is configured to selectively open the fluid collection receptacle **610** for discharge of the urinary fluid **214** collected therein. The cord **640** may interface with the open-close mechanism **638** to effectuate the opening of the fluid collection receptacle **610** and/or may be configured to aid in the manipulation of the fluid collection region **652** of the fluid collection receptacle **610** during use. For example, as the fluid collection receptacle **610** collects the urinary fluid **214**, the fluid collection region **652** may become more cumbersome and awkward to manage. Accordingly, the cord **640** may enable a user to balance the weight of the fluid collection region **652** and/or lift the weight of the fluid collection region **652** to access the discharge port **620**.

[0199] Referring to FIG. 27, the wearable urinary collection system **600** is shown in an exploded view, with the components arranged along a longitudinal axis **654**. During assembly, each of the primary components of system **600** may be translated along this longitudinal axis **654** into their operational positions. For example, the inner sleeve **612** may first be inserted along the axis such that it extends over the shaft **204** of the penis **202**, with the sleeve portion **630** aligned and bands **628** positioned at desired proximal and distal locations to secure the sleeve. Next, the diaphragm

646 may be slid along the inner sleeve **612** and moved longitudinally into engagement with the rim **618**, such that the central aperture **649** forms a seal around the sleeve. The intermediate collection portion **616**, including its double-walled structure and port **636**, may be unfolded or drawn upward along the axis so as to receive the penis glans **206** within its interior.

[0200] If a guide tube **632** is present, it may be inserted longitudinally downward through the port **636** and into the aligned apertures **626** formed within the seals **622**, ensuring that each seal is properly engaged with the tube to create interference fits or valve-like constrictions. The guide tube may be pushed or pulled axially to reach a desired depth that aligns with the lowermost chamber **642**.

[0201] The fluid collection receptacle **610**, including its main fluid collection region **652**, may then be drawn upward from the lower end of the axis until the receptacle encloses the inner sleeve **612** and intermediate collection portion **616**, completing the external containment structure. Once all components are in place, cord **640** may be used to manipulate the receptacle **610** as needed.

[0202] This axial assembly process facilitates efficient alignment and integration of the components along a single continuous direction, thereby simplifying donning and ensuring that internal fluid pathways remain unobstructed during setup.

[0203] FIG. **28** illustrates system **600** being supported relative to the user via dermal elements **656**. These dermal elements **656** may comprise rings, studs, barbell piercings, or other biocompatible anchors that pierce or are implanted into the skin near the base **658** of the penis **202**. The dermal elements **656** may be made of titanium, surgical-grade stainless steel, or other materials commonly used in body modification procedures to reduce the risk of irritation or rejection. The dermal elements **656** may include through-holes, loops, hooks, or magnetic interfaces for attaching tethering components.

[0204] Cords **660** may extend from the dermal elements **656** to the rim **618** or other structural attachment points on the receptacle **610** to distribute weight evenly and support the receptacle **610** during use. The cords **660** may be elastic, non-elastic, or adjustable in length to accommodate user preference and anatomy. In some configurations, tensioning clips or clasps **661** may be used to fine-tune the position and orientation of the receptacle **610**.

[0205] In addition to static support, the cords **660** may also serve to stabilize the vertical orientation of the collection system **600** and preserve axial alignment of the fluid path during user movement. Multiple dermal elements **656** may be used in tandem to reduce point loads and enhance comfort. In alternative embodiments, a flexible harness **662** or wearable belt system may be used in place of, or in conjunction with, dermal elements **656** to suspend the receptacle **610** in a desired position. This configuration may be particularly useful for users unwilling or unable to utilize dermal piercings.

[0206] FIG. **29** illustrates a method **700** for affixing the wearable urinary collection system **600**. The method may include connecting **702** the dermal elements **656** around the base **658** of the penis **202**, thereby establishing fixed anchoring points capable of supporting the weight and position of the fluid collection receptacle **610**. The dermal elements **656** may be secured to the skin using biocompatible piercings or subcutaneous anchors depending on the user's anatomical preference.

[0207] The method **700** may also include inserting **704** the penis **202** through the inner sleeve **612**, such that the sleeve portion **630** extends along the shaft **204**. The elastic bands **628** at each end of the sleeve **612** may constrict gently to maintain longitudinal position, ensuring the sleeve remains aligned without excessive constriction.

[0208] The method **700** may also include positioning **706** the penis glans **206** into the intermediate collection portion **616**, with the urethral opening **208** aligned relative to the port **636**. The glans **206** may rest within the recessed cavity formed by the double-walled structure of the receptacle **610**, directing discharged urine toward the fluid guidance pathway.

[0209] The method **700** may also include positioning **708** the diaphragm **646** of the receptacle **610** along the inner sleeve **612** such that the central aperture **649** elastically engages and seals around

the sleeve **612**. This prevents fluid bypass and reinforces the receptacle's orientation during fluid discharge.

[0210] The method **700** may also include coupling **710** the receptacle **610** to the dermal elements **656** using cords **660**, with each cord extending between the rim **618** of the receptacle and the dermal anchors. The cords **660** may be tensioned or adjusted to maintain an upright orientation of the receptacle **610**, ensuring that fluid travels through the guide tube **632** or equivalent fluid path under gravity without occlusion or pooling.

[0211] While the exemplary implementations of the present disclosure have been described in detail, those familiar with the art will recognize various alternative configurations and implementations within the scope of the following claims. While various implementations may have been described as providing advantages or being preferred over other implementations with respect to one or more desired characteristics, as one skilled in the art is aware, one or more characteristics may be compromised to achieve desired system attributes, which depend on the specific application and implementation. These attributes include, but are not limited to: cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. The implementations discussed herein that are described as less desirable than other implementations or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications. For example, while the several exemplary implementations of the urinary collection receptacle discussed herein are all discussed as being formed of thin, flexible plastic film/sheet material and perhaps also including support rings formed of rigid or semi-rigid materials, the present disclosure is not limited thereto.

Claims

1. A wearable urinary collection system, comprising: a sleeve configured to extend around at least a portion of a penis of a user, the sleeve comprising a guide tube extending from an end of the sleeve; a tubular membrane configured to collect urinary fluid within a volume defined by the tubular membrane, the tubular membrane comprising: a redundant fluid collection portion configured to extend around at least a portion of the sleeve; a series of seals each extending along a width of the tubular membrane, the series of seals spaced along a length of the tubular membrane that is perpendicular to the width, wherein each of the seals includes an aperture; and one or more mating features; and a harness configured to be worn by the user and including a complementary mating feature that is couplable to the one or more mating features of the tubular membrane, wherein the guide tube of the sleeve is configured to extend through the aperture of some of the seals to form a fluid flow path from sleeve through the guide tube and the aperture of remaining ones of the seals into the volume.
2. The wearable urinary collection system of claim 1, wherein the series of seals is configured to inhibit the urinary fluid collected in the volume from traveling to the redundant fluid collection portion.
3. The wearable urinary collection system of claim 2, wherein the series of seals define respective chambers that are each configured to collect the urinary fluid that travels from the volume through the aperture of one or more of the seals.
4. The wearable urinary collection system of claim 3, wherein a terminal end of the guide tube is configured to be disposed in one of the respective chambers.
5. The wearable urinary collection system of claim 2, wherein respective interfaces between the guide tube and the aperture of the some of the seals are configured to inhibit the urinary fluid collected in the volume from traveling to the redundant fluid collection portion.
6. The wearable urinary collection system of claim 5, wherein the respective interfaces between the guide tube and the aperture of the some of the seals are sealed by a constriction.

7. The wearable urinary collection system of claim 1, wherein: the tubular membrane includes an upper portion and a lower portion; the upper portion of the tubular membrane includes the series of seals; and the redundant fluid collection portion is formed from the upper portion of the tubular membrane folding inwards at a rim such that the tubular membrane extends longitudinally from the rim to the lower portion.
8. The wearable urinary collection system of claim 7, wherein the one or more mating features are spaced around the rim of the tubular membrane.
9. A wearable urinary collection apparatus, comprising: an inner member configured to extend around a portion of a penis, the inner member including an end portion configured to extend around a penis glans of the penis, the inner member comprising a tubular portion extending from the end portion; and a fluid collection receptacle comprising: a backflow prevention region comprising a series of redundant collection chambers fluidly connected by respective apertures; an upper region configured to extend around at least a portion of the inner member and fluidly connected to an upper redundant collection chamber from the redundant collection chambers via an upper aperture from the respective apertures; and a urinary collection region configured to collect fluid from the penis and fluidly connected to a lower redundant collection chamber from the redundant collection chambers via a lower aperture from the respective apertures, wherein the tubular portion is configured to extend through the upper redundant collection chamber via the upper aperture to terminate in the backflow prevention region such that the fluid from the penis is directed through the tubular portion and the lower aperture into the urinary collection region.
10. The wearable urinary collection apparatus of claim 9, wherein each of the redundant collection chambers has a width that is larger than a width of the respective apertures.
11. The wearable urinary collection apparatus of claim 9, wherein the upper region includes a rim that defines an upper terminal end of the fluid collection receptacle.
12. The wearable urinary collection apparatus of claim 11, further comprising: a harness configured to support the fluid collection receptacle relative to a user having the penis, the harness including a connecting strap, wherein the fluid collection receptacle further includes apertures spaced around the rim of the fluid collection receptacle that are each configured to receive the connecting strap.
13. The wearable urinary collection apparatus of claim 9, wherein the tubular portion extends away from the end portion of the inner member at a location adjacent to a urethra of the penis.
14. The wearable urinary collection apparatus of claim 9, wherein the upper redundant collection chamber defines a lower surface of the upper region, and wherein the upper aperture extends through the lower surface.
15. The wearable urinary collection apparatus of claim 9, wherein the lower redundant collection chamber is configured to collect the fluid that travels from the urinary collection region through the lower aperture.
16. The wearable urinary collection apparatus of claim 9, wherein the redundant collection chambers are sealed from respective adjacent ones of the redundant collection chambers except through the respective apertures.
17. A method for affixing a urinary collection system to a user having a penis, the method comprising: inserting a penis glans of the penis into a sleeve, the sleeve comprising: an end portion extending around the penis glans; and a guide tube adjacent to a urethra of the penis; inserting the sleeve into a redundant fluid collection portion of a urine collection receptacle, the urine collection receptacle comprising: a volume configured to collect urinary fluid; and a series of sealed chambers disposed between the redundant fluid collection portion and the volume and fluidly connected by respective apertures; and inserting the guide tube of the sleeve through some of the sealed chambers via the respective apertures to form a fluid flow path from the sleeve through the guide tube and the respective apertures of remaining ones of the sealed chambers into the volume, wherein the sealed chambers are configured to collect the urinary fluid that travels through respective interfaces between the guide tube and the respective apertures.

- 18.** The method of claim 17, further comprising: folding an upper portion of the urine collection receptacle that includes the series of sealed chambers inwards at a rim to form the redundant fluid collection portion, wherein the urine collection receptacle extends longitudinally from the rim to a lower portion of the urine collection receptacle.
- 19.** The method of claim 17, further comprising: coupling mating features extending from the urine collection receptacle to complementary mating features disposed along straps of a harness, wherein the straps are configured to extend around respective portions of the user to support the urine collection receptacle relative to the user.
- 20.** The method of claim 17, wherein inserting the guide tube of the sleeve through some of the sealed chambers via the respective apertures forms respective interfaces between the guide tube and the respective apertures that inhibit the urinary fluid from traveling from the volume to the redundant fluid collection portion.
-